So You Want To...

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Chapter 1

Welcome to SYWT!

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1.1 So You Want To... Get An Introduction?

1.1.1 Know Why This Document Exists?

So you've decided to pursue research in Physics? Congratulations! You are embarking on a fantastic adventure of discovery into the fundamental laws of nature, as well as on a journey of self discovery. Through your investigations into physics, you will gain a wide range of skills relevant for a research-oriented career. These skills will serve you well whether you decide to pursue a job in academia or pursue research outside of the ivory tower. You are fortunate to have the opportunity to push the boundaries of scientific knowledge and grow yourself in innumerable ways.

If you are reading this document, then you are likely a current or former member of my (i.e. Jesse Thaler's) research group. Or perhaps you are a perspective group member who wants to get a sense of my advising style. I have been a faculty member at MIT since 2010, and I have worked with 25 B.Sc. students, 3 M.Eng. students, 13 Ph.D. students, and 19 postdocs as of January 2026. I was also an (under)graduate student and postdoc myself, and I wish someone had told me all of the things that I have learned via experience and osmosis. The goal of this document is to provide you with information to help guide you on your research journey and pass down some institutional knowledge to the next generation of aspiring physicists.

One of the key lessons I have learned over the years is that **there are as many ways to be a successful physicist as there are successful physicists!** There is no one right way "to physics", just as there is no one right way "to academia", and certainly no one right way "to professor". But I have also learned that there are common goals and experiences that almost all physics researchers face. This document has lots of information in it that I wish I had been told on day 1... and information that I find helpful to remind myself about on day $\mathcal{O}(10{,}000)$.

1.1.2 Know How This Document Is Structured?

As you've undoubtedly figured out already, this document is structured as a series of "So You Want To...?" (SYWT) questions. At some level, this structure is just a gimmick. But looking back on my experience as a student, postdoc, and now professor, I have realized that the pathway to success (however you choose to define it) is paved with specific well-defined tasks. It is daunting to try to tackle complex and multifaceted goals, e.g. "So You Want to Be a Rock Star?". It is much easier to identify smaller sub-tasks along the way, e.g. "So You Want To Learn How to Hold a Microphone?". The questions in this document are ones that I have been asked so many times that I felt a responsibility to write them all down. The organization of this document is an attempt to arrange the questions with some logical coherence.

There is a lot of content in this document! Much of this document is structured according to career stage, with answers to the key questions most physicists face at the corresponding point along their journey. In the appendices, there is additional advice relevant for researchers at multiple career stages. This document will be revised periodically as we learn new things together.

What you will not find, though, is the answer to life the universe and everything. The path to being a successful physics researcher is more than just following the advice of your elders, though that is a reasonable place to start. You will need to keep asking questions and being curious in order to thrive, no matter your career stage. You will need to set your own goals and decide what is best for you (and your family). And just as this document will evolve, your goals will evolve month to month and year to year. Your journey has just begun... and in many ways, it will never end.

1.1.3 Know Who Contributed to This Document?

This documented was authored primarily by me (Jesse Thaler), but I received advice, suggestions, and contributions from many current, former, and honorary members of my research group, including:

- **Samuel Alipour-fard:** Special thanks for the template file and contributions to Sec. A.2.
- **Sean Benevedes.**
- **Cari Cesarotti.**
- **Rikab Gambhir:** Special thanks for contributions to Sec. C.1.
- **Patrick Komiske.**
- **Lina Necib:** Special thanks for contributions to Sec. A.4.
- **Pedro Rivera-Cardona.**
- **Stella Schindler.** Special thanks for contributions to Sec. 4.4.
- **Sokratis Trifinopoulos.**
- **Octavio Vega.**
- **Ray Wynne.**

Additional advice directly from my group members is provided in Sec. B.

Thanks to Bob Hewes for the original idea of trying to do "advising at scale" through a written, evolving document. Thanks to Ana Gabela for help with editing, organization, project management, and cheerleading.

1.1.4 Improve This Document?

Thanks for offering! I would value your help and ideas! This is a living document that will grow/shrink/morph as needed. Please fill out this form with your suggestions for edits and/or content to add.

1.1.5 Understand This Document's Version Number?

This document is labeled with a date and a version, currently February 19, 2026 (v2.1.6). The date should be self explanatory. For the version, I am trying to follow the recommendations of Semantic Versioning, where the version number MAJOR.MINOR.PATCH corresponds to the degree of changes I have made.

- MAJOR: I am currently on v2 of this document. The original v1 was based on crowd-sourcing information, but it because clear that such an approach would not lead to coherent advice. Hence the major change in v2 to single authorship, with advice from my group members (mostly in their own words) in App. B. I plan to increase the major version number when I add or substantially revise a whole chapter.
- MINOR: Any time I add one or more new substantive sections, or fundamentally change my advice in an existing section, I'll increase the minor version number. My aspirational goal (not yet realized) is to increase the minor version number once a month.

- PATCH: Increasing the patch version number is mostly for fixing typos, but I'll also use it when I revise a section or improve the formatting without fundamentally changing my advice.

1.1.6 Influence What Jesse Writes Up Next?

Writing this document is part of my journey into improving my advising/mentoring. For parts in development, I've used the (∄) symbol to indicate topics that do not (yet) exist or need substantial revisions.

Here is my rough priority for the next things I plan to write up in SYWT as of January 2026:

- **Advice for postdocs.** Sec. 4 audience: postdoc who just came to MIT.
- **Advice for junior faculty.** Sec. 5 audience: faculty member who just came to MIT.
- **Giving a talk.** Sec. D.2
- **Writing a paper.** Sec. D.1.
- **Advice for undergrads.** Sec. 2 audience: a student who is thinking about doing a UROP in my group.
- **LaTeX warnings.** I am getting $O(100)$ LaTeX warnings related to the table of contents and hyperreferences. If you encounter a link that doesn't work, let me know, since that would help me debug what's going on.

That said, I am open to adjusting my priorities if there is something pressing you need advice on.

1.2 So You Want To... Work in Academia?

1.2.1 Anticipate the Academic Trajectory?

If you are reading this document, then you probably have "MIT" somewhere on your cirriculum vitae. Assuming that's the case, then you will have every advantage in the academic job market. Yes, there are horror stories of people who are well-respected (and well-cited) but don't get tenure-track positions. Yes, the competition for jobs at the most selective universities is fierce. Yes, you may face difficult trade-offs between personal and professional goals. Yes, fluctuations in federal research funds and research priorities can dramatically affect the availability of academic jobs. But you have already demonstrated your research potential by getting into MIT, and you will be (or have already been) trained well during your time here.

There is a pervasive myth among non-academics that being a faculty member at a research university is somehow a cushy job. My mother, who worked as a high school educator, still needles me for "only" having to teach one class a semester. When I visit high school classrooms for outreach events, the students are shocked to find out that I work (hard!) over the summer. When I go on faculty sabbatical every seventh year, people I meet are surprised to find out that I am "still" doing research (and loving it). In many ways, being a faculty member is less like being an educator and more like being an entrepreneur, CEO, or professional athlete. This is part of the reason why I love my job, since it is dynamic and challenging. But it is also part of the reason why academia is not for everyone.

If you do decide to spend a lifetime in academia, here is what you have to look forward to:

- **Undergraduate school (typically 4 years).** See Sec. 2. You will be introduced to the foundations of physics through your courses, but the real action happens when you start doing research. If you are enthralled by solving research problems, then you owe it to yourself to apply to...
- **Graduate school (typically 5 years).** See Sec. 3. Once you get your coursework out of the way and pass your oral quals, you will be building your research portfolio. The actual PhD thesis document is kind of a joke; what really matters are your publications. You will want to have at least three papers (and typically double that) where you played a essential role. A strong research portfolio will put you in a good position to land...
- **Your first postdoc (typically 3 years).** See Sec. 4. Essentially no one gets a faculty job right out of grad school. (And even if they do, they typically negotiate hard for a deferral to do a postdoc.) Your first postdoc is a chance to demonstrate your independence and develop a compelling research vision. An innovative research program will put you in a good position to land...
- **Your second postdoc (typically 3 years).** Re-read Sec. 4. Wait, what? One postdoc isn't enough to land a faculty job? In some cases, you do need more time to build up your research program. But sometimes you are just unlucky and there aren't faculty openings in your area or your work is too new to get the attention of a faculty hiring committee. But if you stick it out for your second (or third or fourth) postdoc, you might land...
- **A tenure-track faculty position (typically 7 years).** See Sec. 5. Now the fun really begins, because you have new responsibilities in addition to your research. You have to secure grant funding, teach classes, mentor students, and perform service work. If you can do all of these things while building a successful research group, you might get...
- **Tenure!** The promised land of academic freedom! Pop the champagne! As one of my colleagues put it: "You survived the rat race, and now you realize... you're a rat." Why did you make all of those sacrifices over the

years? What is your purpose in life? Does your research have any value to society? If you can make it through the post-tenure depression, then you might achieve...

- **A fulfilling academic career (typically 25 years).** See Sec. 6. Maybe you reinvent your research program. Maybe you go into academic leadership. Maybe you start a company or run for political office. The academic path has the potential to open exciting new doors, if you want to explore. Or maybe you just keep your head down and keep doing your research until you...
- **Retire.**

Sounds fun, right? Did I mention that each of the career stages above typically involves moving to a different geographic area? Did I mention that the time of core research productivity also overlaps with typical child-bearing years? Did I mention that you will see your friends who left academia have fulfilling careers with better work-life balance (and often better compensation)? When I was considering going to physics graduate school, a family friend (and physics PhD) tried to dissuade me from the academic path for all of these reasons and more. See Sec. 3.4.1 for further things to consider.

For me, though, I can't imagine doing anything other than curiosity-driven research. The university environment is basically the only place I can pursue this dream. (There are also similar roles at national labs and some companies, but they essentially operate just like academic positions and are just as competitive.) I also enjoy the multi-faceted nature of being a professor, even if I look back at my research-focused grad school and postdoc years with envy at my former self. If you want an academic career, there is a reasonably high probability you could land a professorship if you are willing to make some level of personal or professional sacrifice. That said, don't just follow my dream since you think you have to. Do it because the academic career path energizes you and the physics brings you joy.

1.2.2 Really Know What You Signed Up For?

I am a tenured faculty member, and my memories have undoubtedly been softened by nostalgia (and survivorship bias). This document is written by someone who found academic science to be a welcoming and energizing community. But not everyone is going to have the same (positive) experience in the physics community as I did.

Sadly, it is not all sunshine and roses out there, especially if you are member of an underrepresented group. You may encounter discrimination and harassment. You will certainly encounter well-meaning people who nevertheless make you cringe. Some advice for dealing with these kinds of situations is given in Sec. A.4. If you are experiencing an immediate crisis, ask for help or call 911.

Improving the culture of academia is something that I care deeply about, and I hope I can make a meaningful difference in my lifetime. I don't want to ignore the negative aspects of academia, but I sometimes struggle to articulate my perspective in writing. Eventually, I hope to have concrete SYWT action items for enacting change. If you encounter discriminatory behavior, or if you just want to talk about these issues, please reach out to me.

1.2.3 Cultivate Your Curiosity?

Most of this document is about navigating the wonderful but esoteric world of academic research. In reading back over my advice, though, I realize that I have assumed that you are fundamentally interested in **cultivating your curiosity**. You have questions about how the Universe works, and an academic path offers you an unparalleled opportunity to get answers, or pieces of answers, or at least empower you to ask more refined questions. That is to say, no matter how good

you might get at handling the logistical challenges of academia, you will only have a fulfilling academic career if you pursue the questions that flow from (and feed into) your curiosity.

How can you hold on to your child-like curiosity and avoid falling into the trap of adult-like cynicism? I wish I had a good answer for you. When my brother passed away in 2011 (see App. A.2.4), he wrote his own eulogy which included this piece of sage advice for me:

- **Don't let university bureaucracy distract you from your creative process.**

My brother didn't tell me how I was supposed to avoid the bureaucratic brambles, but he nevertheless gave me a vivid reminder of my core motivations.

One day, perhaps I could develop a "curriculum of curiosity" and teach students how to ask the kinds of questions that can fuel a lifetime of physics research. While I don't necessarily think that curiosity can be taught, I do think that starting from a seed of naive wonder, one can cultivate a tree of professional expertise, with ever-growing branches that reach upwards and extend outwards. But it all starts from that seed, so make sure to choose a research path that aligns with the things that you are intensely curiosity about. If you are just looking for a job (or a job title), there are much more straightforward paths than the academic one.

1.3 So You Want To... Continue Your Journey?

I hope you find this document to be a valuable source of information about the academic physics experience. Are you sold on joining my research group? Great! Does my style rub you completely the wrong way? Also great, since it is better to learn this now rather than later.

I recommend starting with the chapter corresponding to your career stage:

- **Undergraduate Students**: Sec. 2
- **Graduate Students**: Sec. 3
- **Postdocs**: Sec. 4
- **Junior Faculty**: Sec. 5
- **Tenured Faculty**: Sec. 6

Then you should dig into the appendices, which are overflowing with helpful information.

- **General Advice from Me**: Sec. A
- **Advice from My Group Members**: Sec. B
- **Academic Jobs**: Sec. C
- **Research Skills**: Sec. D
- **Teaching Skills**: Sec. E
- **Mentor/Mentee Skills**: Sec. F
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And do give me feedback on ways to make this document more helpful. Onward!

Chapter 2

Advice for Undergraduate Students

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2.1 So You Want To... Get Advice as an Undergraduate Student?

When writing this chapter, I imagined handing it to an MIT physics major who is interested in pursuing a UROP project with me. For undergraduates potentially interested in going to graduate school, the summer between their junior and senior years marks a key decision point. I strongly recommend students pursue research during that summer to see if physics research is something they might enjoy doing as a career. Of course, it is possible to get involved in research sooner than that, but certainly in my case, the junior-to-senior summer was essential for defining my academic trajectory.

2.1.1 Know What You Signed Up For?

You signed up to get a Physics degree. You signed up to learn about the foundations of modern physics, from quantum mechanics to relativity. You signed up to better understand the inner workings of the universe through the dual lenses of mathematics and empirical observations. You are learning that the underpinnings of nature are simultaneously simpler and more complex than you could have ever imagined.

Many of you signed up to get your first taste of physics research. In the classroom, you learn what is already known, and you are rewarded for being clever, careful, and fast. In research, you are working on the edge of human knowledge, and you are rewarded for being curious, critical, and patient. There is no guarantee of success in research, nor is there one definition of "success" for every researcher.

At the end of your time as a undergraduate, you have a key design to make. Do you want to go to graduate school in Physics (or a related field) and explore deeper down the rabbit hole? Or do you want to apply the abstract-reasoning and problem-solving skills you gained as a Physics major to tackle different kinds of challenges. The best way to make this decision is to immerse yourself in physics research, to see if you want to embark on a longer voyage of discovery.

2.2 So You Want To... Work with Jesse? (♯)

2.2.1 Know the Prerequisites for Theory Work?

For undergraduates who are thriving in the Physics major, the next natural step is doing a physics research project. There is a huge range of different kinds of physics research, from experimental fabrication, to data analysis, to theory calculations. I highly recommend all Physics majors try doing a research project sometime during their undergraduate years, especially if they plan on eventually applying to Physics graduate school. Research can be a full-time job, so you'll get the most out of your research experience by invested substantial time in it.

Here are some things to consider when deciding if you have the necessary background to do a theoretical research project with me.

- **The absolute minimum preparation for theoretical physics research in my group is special relativity and quantum mechanics.** These are the two pillars of modern physics, and for my research related to particle physics, it is basically impossible to doing anything meaningful unless you already have some familiarity with these topics. At MIT, you'll want to a take a course on **Relativity (8.033)** and a full year of **Quantum Mechanics (8.05/8.06)**, otherwise you'll struggle with basic things like manipulating four-vectors or estimating cross sections. In some cases, it is possible to start get involved in more computational research without those prerequisites, but you should nevertheless try to get these foundations as soon as possible.
- **Physics coursework and physics research are completely different activities.** Many students are drawn to theoretical research because they think it will be similar to the coursework they have already done (and enjoyed). There is a huge difference, though, between problem sets in courses and the problems you encounter in research. Students who are used to getting all of the answers right on an exam will be frustrated to learn that research questions are far less well defined, and therefore much harder to answer. Sometimes I've found that students who have struggled with their classes actually do better in research, since they are used to dealing with setbacks.
- **Theoretical research can be just as "dirty" as experimental research.** Many students are drawn to theoretical research because they think it will be more elegant than experimental research. Remember, though, that physics is ultimately an experimental science. So even if you don't work directly on an experiment, if you work with me, you will be doing research adjacent to experimental realities. This means actually needing to compute numbers, not just deriving symbolic expressions. It also typicall means making lots and lots of plots.
- **These days, most theoretical research requires strong computer skills.** There are very few meaningful pencil-and-paper calculations one can do in theoretical physics research, at least in my group. Even though our research is based on first principles from quantum field theory, and even though many of our discussions will happen at the blackboard, we end up doing a lot of coding and/or data analysis work. So while not a strict prerequisite, some experience with scientific computing is recommended to make the most out of working in my group.

These last two bullet points mean that undergraduate students who work with me on theory projects often find that their day-to-day activities are similar to their friends working on data analysis for particle physics experiments. So if you are considering working with me, also consider working with one of my experimental colleagues. The culture of "theory" and "experiment" is certainly different, and no one would accuse me of being an experimentalist. But if you are looking to do more mathematical physics, with no connection to experiment, I'm probably not the best choice of research advisor.

2.2.2 Choose an Initial Research Direction? (ㄱ)

2.2.3 Prepare for Your First Few Meetings with Me? (ㄱ)

2.3 So You Want To... Get a Roadmap for the Physics Major? (∄)

2.3.1 Navigate Your First Year? (∄)

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2.3.3 Focus During Your Junior Year? (∄)

2.3.4 Take Flight During Your Senior Year? (∄)

2.4 So You Want To... Go to Physics Grad School? (♯)

2.4.1 Organize Your Grad School Search? (♯)

2.4.2 Decide which PhD Programs to Apply to?

Especially for students who come from MIT, there is a tendency to think that they should only apply to "the best" schools. Certainly, one should apply to ones top choices, but it would be a mistake to think that good science happens at only a small number of elite institutions. In general, the quality of faculty has a longer tail than you might extrapolate from the selectivity of schools at the undergraduate level. This makes some intuitive sense, since the proportion of qualified faculty applicants is relatively high compared to the number of available tenure-track positions, whereas the reverse tends to be true for undergrad slots.

What this means in practice is that you can more or less guarantee your admission to Physics grad school if you apply to a range of schools, including those that might not have been on your radar beforehand. Remember, you only need one great PhD advisor to have a great PhD experience, though of course having a vibrant PhD community is definitely important as well and should factor into your decision. But you can launch a successful career from many different starting points.

Here are some general recommendations for making your list of schools:

- Remember that **grad school admissions is a probabilistic process**. For reference, MIT receives roughly a factor of 3 more highly qualified applicants than slots available in the program (and even more applicants in total). Even if you do everything right, there are random elements beyond your control, so you will need to play the odds.
- Plan to **apply to around 9–15 schools in total**. Fewer than that, and you risk not getting in anywhere. More than that, and you probably haven't done your homework to determine which schools would be a good fit to your interests. One exception is if you are applying to different types of PhD programs (e.g. simultaneously in Physics and another field), where you might need to expand your search to keep your options open.
- Always apply to your top choices, but **around half of your list should consist of less selective institutions**. Note that less selective means easier to get into, not lower faculty quality. Once you start researching schools, you'll find great faculty at a range of places.
- For any school you put on your list, you should **explicitly identify 1-3 faculty members** whose research profile excites you. Some schools (even top schools) might not have invested in your preferred research area. If you aren't sure what you want to do, make sure there is a range of faculty members that span your potential interests. See App. C.4.1 for a starting point for possible schools.
- If there are particular papers that have influenced your work, **identity the institutions of the authors**. Your future PhD advisor might already be someone you know from their work! This is also a good way to see that top researchers have a wide range of institutional affiliations.
- **Ask current MIT graduate students where they applied** and what criteria they used to select schools. In some ways, current grad students are like a "time machine" to see where you might end up in a few years.
- **Ask faculty for advice**, but only after you've made a preliminary list. Faculty can help you identify places you might have missed, but they can also have blind spots, so you'll want broad input from others as well.

- The Theoretical Particle Physics Jobs Rumor Mill is a great place to **identify schools that have hired junior faculty** in the past 5-10 years. Having relatively recent hires is usually a good sign about the vibrancy of a research group and the level of institutional investment in theoretical particle physics.
- Take a look at this plot showing where recent faculty hires got their PhDs. Long bars correspond to the "usual suspects", but you can also see the long tail effect, where **recent hires got their PhD at a wide range of schools**. (Of course, if you don't plan to continue on in academia after your PhD, this information will be less relevant to you.)

2.4.3 Choose a Physics PhD Program? (♯)

2.4.4 Decide Between Theoretical and Experimental Research? (♯)

Chapter 3

Advice for Graduate Students

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3.1 So You Want To... Get Advice as a Graduate Student?

When writing this chapter, I imagined handing it to a first-year MIT graduate student who is trying to decide whether to join my research group. While spots in my research group are constrained by resources (i.e. my time and the availability of funding), a much bigger factor is whether my research style is a good match to yours.

3.1.1 Know What You Signed Up For?

You signed up to get a PhD. You signed up to transform yourself from a physics student into a physics researcher. You signed up to gain a level of problem-solving skills and independence that is difficult to obtain without pushing against the boundaries of human knowledge. You signed up to pursue your physics passions and see how far down the rabbit hole goes. Earning a PhD will likely be one of the hardest things you achieve in your lifetime.

Many of you signed up to get on the road to an academic career (see Sec. 1.2.1), either toward a professorship or an equivalent position at a national laboratory, non-profit foundation, or for-profit company. In my opinion, being a physics researcher is the best job in the world. Being a professor, on the other hand, is a complex multi-faceted endeavor that involves research as just one component. For me, it is still the best job in the world, but it involves trade-offs and sacrifices that might not be apparent to you yet.

As you pursue your PhD, remember that there are many paths to a research career outside of academia. Your degree will be immensely valuable and a source of personal satisfaction no matter what career path you ultimately choose. There are fantastic research opportunities in a range of jobs that benefit (directly or indirectly) from a physics background. But if you decide to continue in academia, you will have the opportunity to engage in curiosity-driven research, which is a joy, a gift, and a responsibility.

3.2 So You Want To... Work With Jesse?

3.2.1 Join Jesse's Research Group?

Choosing a research advisor is one of the most important decisions you will make as a graduate student. (It also important for undergraduate students and postdocs, but less so.) Your research advisor will be the person you will turn to most often for advice and your primary advocate for awards and jobs. And for better or worse, some of the quirks of your research advisor will rub off on you.

You should explore as many options as you can when choosing a research advisor, both before and after you select a graduate program. Here are some questions to consider when deciding whether or not to work with me:

- **Am I a good fit to your research interests?** I have worked broadly across a number of domains of theoretical particle physics, and I tend to be open-minded about pursuing new directions. What I'm working on now may not be what I'm working on 2 years from now, though my changes tend to be adiabatic. So you should think about how much flexibility of direction matters to you and how much you are interested in driving your own research direction.
- **Am I a good fit to your research style?** I expect my students to be self-motivated and striving towards becoming more independent. This means that I tend to treat research projects less as tasks to be completed and more like opportunities to explore my interests and grow as a researcher. The question I start working on is rarely the same as the question I ultimately write a paper about. This flexibility is one of the joys of research for me, but it may be challenging for you if want more concrete research projects.
- **Are you willing to be your own advocate?** I expect my students to set their own agenda, both in research and beyond. Because you have the best understanding of your goals, you are in the best position to advocate for yourself and your research vision. Be assertive about your goals and needs, and I will try to give you the best advice and guidance I can. Of course, sometimes that advice will be to seek guidance from someone other than me.
- **Are you willing to proactively communicate?** I wear many hats and have (too) many commitments. Research, though, is my top priority, and I will always make time to discuss research projects with you. Because my schedule is so packed, though, you will usually need to initiate communication. If you have a great research result, please share! If you need help on something, please ask! In my experience, a lack of communication is often the first sign of a problem. So whether you set up an in-person meeting or just send a quick message over Slack, keeping me informed is the best way to keep me engaged. I tend to "**impedance match**", meaning that if I see that you are making progress on a project, I will try to catch up as best as I can.
- **Have you asked my current/former students about me?** Although not always the case, past behavior is typically a good prediction of future behavior. Talk to people who have worked with me in the past and figure out whether I was an effective advisor for them. Anecdotes aren't data, but they are the best you have when you are talking about a small numbers game.

Here are some reasons why students have decided <u>not</u> to work with me:

- **They would prefer to work on more formal topics.** My research tends to be quite phenomenological, and increasingly computational. While I enjoy blue sky thinking, I tend not to stray too far from experimentally testable ideas. There are exceptions, like my research on more formal aspects of supersymmetry, but I tend to

expect my students to be comfortable calculating actual numbers (with units!). Often this means being comfortable with coding and numerics.

- **They want more direction from an advisor.** I always have a long list of potential projects, but I've found that it works best when a student has a rough research direction for themselves in mind. I can certainly help in identifying projects, but I'm not good at telling students exactly what to do.
- **They find the way I engage to be too high energy.** Yes, I am a high-energy theorist, in two meanings of that phrase. I engage deeply in conversations and I tend to ask a lot of probing questions, which is my way of learning and understanding. One my peers said that working with my PhD advisor "was like having Usain Bolt as a track coach". While I'm not operating quite at that level, I am passionate and dedicated and I expect my students to aspire to the same.

Here are some reasons why students have decided that they do want to work with me:

- **They want honest feedback coupled with constructive suggestions.** When something isn't right, I say so. When something is great, I (try to remember to) say so. I have high standards, and I am always ready to brainstorm ways to help you reach that standard.
- **They want to engage with me as a collaborator.** I value independence, and I want you to come up with your own ideas and approaches. Of course, there will always be a mentor/mentee dynamic to some extent, and I have a habit of quizzing students at the blackboard. But for research, I prefer working in a collaborative mode whenever possible.
- **They care about research relevance to the "real" world.** I'm still a theoretical physicist at heart, but I get the most professional satisfaction when my work has an impact on experimental (particle) physics.
- **Someone told them to work with me.** If a trusted adviser suggests to you that I could be your next trusted advisor, that's advice you should give at least some weight to.

Luckily, there are many different faculty at MIT and in the CTP with different research interests/styles. So if I'm not the right fit for you, find someone who is!

3.2.2 Leave Jesse's Research Group?

This is kind of a funny question to have in a document like this, so let me first explain why I'm including it. Choosing your research home is one of the most important decisions you can make as a student, and you don't want the sole reason to stay in a group to be because of inertia or a sense of duty. One of the best decisions I made when I was graduate student was to "quit" string theory, and I want to make sure that the members of my group have the freedom to choose their own path, whether they continue to work with me or not.

In that spirit, here are some questions to consider if you are contemplating leaving my group:

- **Does your research style match mine?** I have a particular research style that may or may not work with your style. That's okay! It might be that our styles complement each other, but it might not. You want to find a research home that challenges and excites you, but also makes you feel comfortable. It might be that you would be more satisfied in a different research group, and I totally get that.

- **Have you identified greener pastures?** There are many exciting research opportunity in the CTP and elsewhere at MIT (and in the Boston area and beyond). If you find something more exciting to work on in another group (or find an advisor with a style that better matches yours), I will not only be understanding but also be encouraging of you to follow your interests (and style).
- **Do you want to expand your collaborator pool?** You don't have to leave my research group to collaborate with other researchers! A number of my students have pursued outside collaborations. In fact, I encourage outside collaborations by my students, since I think they are essential for their professional development. At the same time, if your primary research interests are significantly different from mine, then it might make sense to find a different advisor.

If you are on the fence about leaving my group, here are some additional things to consider:

- **I will support you no matter what decision you make about your future.** I would hope that you would give me ample notice before leaving my group (and wrap up loose ends for ongoing projects). But you can "break up" with me without my permission. Sometimes you just need to leave to find the right path to follow.
- **You may feel better if you involve a third party.** Having difficult conversations is always challenging, and you should invite a trusted third party to a meeting with me if that would make you feel more comfortable, no matter what topic you want to discuss.

Finding a comfortable research home is essential to research success, so I encourage you to seek out multiple perspectives as you weigh your advisor decision.

3.2.3 Plan for the First Few Meetings With Jesse?

Each student I have worked with has had a different onboarding experience. Sometimes we jump right into research, sometimes there are preliminary papers to read, sometimes a student needs to focus on their classes, etc. Because each student has a different background coming in, I do not have a fixed set of expectations at the outset.

My goal during our first days/weeks/months together is to give you a sense of my research and advising style. I will undoubtedly ask you a lot of questions in our individual meetings. These questions will be about what your goals are for the PhD, what topics you want to work on, and what you already know about the field. You should also come to our weekly group meetings, so you can get a sense of my group's culture.

Your goal during our first days/weeks/months together is to figure out if I'm a good advisor fit for you. Part of this is being assertive about what you want. I am open to learning new ways to interact with my advisees, so if you tell me what kinds of guidance work best for you, then I might be able to adapt.

3.2.4 Know What Jesse Expects of You?

I have very few hard and fast rules for the members of my research group. But I do have some basic expectations for your graduate school experience.

- **Uphold the MIT Physics Community Values.** Two of my former students were heavily involved in authoring the MIT Physics Community Values. I expect you to uphold these values and help others uphold them as well. The basic principle behind the values is that:

  - **Members of our community should treat each other with respect and decency at all times. In turn, we should not alienate, diminish or insult each other, either in word or deed.**

  The five specific values are:

  - **Well-being**: We support each other at all times and remember that we are not alone.
  - **Respect**: We value the multitude of ways to be a physicist and the many paths through our field and Department.
  - **Inclusion**: We strive to speak and act in ways that support and include all members of our community.
  - **Collaboration**: Physics is a social endeavor and we proudly collaborate with others to advance the field.
  - **Mentorship**: All physicists are here because of the mentorship we have received and continue to receive, and the mentorship we offer to others.

  More information about the values is available from the Physics Values Committee.

- **Set personal and professional goals.** At the beginning of each semester, I will ask you to write down your goals and share them with me. At every career stage, it is important to understand what your personal and professional goals are and put them into writing. This is a way to hold yourself accountable, and it will help you structure your time and attention to achieve those goals. By sharing the goals with me, we can check together that the way you are allocating your time is aligned with the things you hope to accomplish. Goal setting is also a two-way street; see Sec. 3.2.6.

- **Communicate regularly with me.** Due to my responsibilities as IAIFI Director, I cannot commit to a regular meeting slot with each of my students. But you should nevertheless aim to engage with me every week or every other week, separate from group meetings. For an in-person meeting, you can always schedule yourself on my calendar (schedule two blocks if you need an hour). And I respond relatively quickly to email and Slack messages. Sometimes we'll need to meet more often if we are trying to get a paper out, and sometimes we won't need to meet as often if you are making steady progress. But if I don't hear from you regularly (either virtually or in person), then I won't be able to give you guidance.

- **Ask questions if you are unsure.** There are many unwritten rules in academia (though I'm trying to write many of them down in this document!). If you are unsure about how something is supposed to work, please ask me. Even if something seems obvious to you, it is better to ask. I have never been upset about someone asking me a question, but I have been upset when something was done incorrectly without consulting me.

- **Observe the 72 hour rule.** If you are banging your head against the wall for 3 days and not making research progress, then it is time to talk to me, or at least send me a Slack message. If you've only struggled with a problem for a day, then you'll often learn more by trying to figure it out yourself or by talking about it with your friends. But after 72 hours of zero progress, then you've reached the point of diminishing returns. You can certainly reach out to me before 72 hours, since I love answering questions and debugging problems, though I might not prioritize your problem until the 72 hour mark.

- **Treat graduate school as a full-time position.** I hesitate to use the word "job", since you are working for yourself much more so than you are working for me. But it is useful to calibrate your effort to a full time job (nominally 40 hours a week). It is important to take vacations and have hobbies, and I don't expect you to respond to emails

outside of usual work hours. But I do expect you devote substantial time and energy to your graduate research. Ideally, you will be intrinsically motivated to put in the time to get your PhD. Given the flexibility of academia, I understand that your effort will naturally fluctuate day by day and week by week.

- **Balance research with your other responsibilities.** Getting a PhD is primarily about doing great research, so you'll want to devote substantial time to your research each (non-travel, non-vacation) week. You will have other important responsibilities, like taking classes, being a teaching assistant, and serving on committees. And as you get further on in your PhD program, you should look for opportunities to mentor your junior colleagues. But make sure to protect your research time, since that is your primary role. (Conversely, if you primarily enjoy the non-research aspects of the PhD program, you should talk with me about alternative paths forward.)
- **Attend my weekly group meetings and key events/seminars/colloquia.** You have the flexibility to set your working hours, but you should participate in the life of my group/CTP/IAIFI by attending all key events, unless you are away from MIT. Here is an example talk schedule for one of my students:
  - Monday Group Meeting (of course!);
  - Monday Nuclear/Particle Seminar;
  - Monday LNS Colloquium;
  - Thursday IAIFI Lunch Discussion;
  - Thursday Physics Colloquium;
  - Friday CTP Grad Student Lunch; and
  - Friday IAIFI Events.

  Yes, that's a lot of time each week spent in talks! But this is the best way to stay up to date about what's going on in the broader field, and these events often spark new collaborations and research directions. Of course, the specific talks you attend will be determined by your research interests.
- **Help me improve as an advisor.** I try to do my best, but I have my blind spots. If there are ways I could be a more effective advisor, tell me directly or tell someone whom you trust to tell me. There are some aspects of my advising style that I might be reluctant/unable to change, and in some cases, you would be better off switching advisors (see Sec. 3.2.2). But I try to be open to suggestions, within reason.

3.2.5 Know What You Should Expect of Yourself?

There is no one way to be a grad student, but there are generic things you should expect from yourself during your time getting a PhD.

- **Prioritize your mental and physical health.** You can't do your best research if your brain and body are deteriorating. Remember that wellness has a daily/weekly component (e.g. sleep, mindfulness, exercise) and a monthly/yearly component (e.g. relationships, vacations).
- **Have a growth mindset.** The academic timeline tends to be broken into ~ 4 year chunks. Think about how much you evolved during high school and during your undergraduate studies; you will experience that same level of

growth as a graduate student. In fact, you will likely experience more growth per unit time than in your past, since you will be setting your own goals, which are often loftier than the ones set in the standard classroom environment.

- **Be patient with yourself.** It can sometimes be hard to see how much you are improving when you are in the thick of it. Sometimes you get stuck in research for extended periods of time with little tangible progress. But if you keep showing up and keep putting in effort, you will make it out of your slump eventually, so be patient. Every one of my PhD students has gone through a phase transition where in span of a relatively short time, they somehow "get it". But like satori, it is difficult to know what "it" is or when it will happen, until you get there.
- **Push yourself out of your comfort zone.** For some students, getting out of their comfort zone means working on their verbal and written communication skills. For other students, it means learning how to code or how to sit down and pound out a complex calculation. Whatever it is that you procrastinate the most about is the thing you will have to work the hardest at, so now is the time to start pushing your boundaries.
- **Aim to become self directed.** At the beginning of your PhD, you will undoubtedly need my help to figure out how to pursue your research goals. But as you advance, you will want to take more and more control over what you do and how you do it. One way to think about this is that you want to grow beyond being my student and becoming my collaborator.
- **Be humble.** Imposter syndrome is real, and academics often underestimate their abilities. But hubris is also real, especially among theoretical physicists. Acknowledge what you know, but also what you don't know.
- **Maintain your curiosity.** As you become more specialized in your research, there will be growing incentives to narrow your focus. Remember why you became a physicist in the first place and maintain broad interests beyond your immediate horizons. Oftentimes, key research insights can arise just by being curious about developments in fields adjacent to yours. See Sec. 1.2.3 for further thoughts along these lines.
- **Regularly seek advice.** There are a lot of people (besides me) who have "been there, done that." Ask questions of your peers and your elders, and start building your mentoring network.
- **Invest in the community.** There is a high likelihood that some of the people you meet in graduate school will become your lifelong colleagues, collaborators, and friends. The people you learn from now might become the people you lean on professionally. So invest some of your time in building the kind of community that you want to be a part of.

3.2.6 Know What to Expect of Jesse?

I care deeply about the well-being and success of my mentees. My students are my top priority research-wise, and I will strive to make sure you have access to as many opportunities as possible to advance your research career. This means that I am always willing to read drafts of manuscripts, watch practice talks, write letters of recommendation, provide advice, etc.

What I cannot do, though, is provide extensive counseling or tutoring. To get a PhD, you will need to build up a network of friends, collaborators, mentors, and supporters. I will of course be a key person in that network and the one most responsible for helping you shape your research portfolio. But when you defend your PhD and acknowledge all of the people who helped you along your journey, I will be just one of many people invested in your success.

We will formalize our expectations of each other at the beginning of each semester. Along with asking you to set your personal and professional goals, we will brainstorm the best way for me to support you. Of course, we can make adjustments throughout the semester, and you should feel empowered to "manage up" and help me help you.

If there is a breakdown in communication and I am not following through on the things I said I would do, there are various informal mechanisms for intervention. One is talking to your academic advisor. Another is talking to a person in a leadership role in the Physics department, such as the CTP Director, Graduate Student Coordinator, or Graduate Student Advocate. If you have formed your thesis committee, you can reach out to your thesis committee members. You can also talk to other graduate students and postdocs in my group for their advice.

In terms of formal mechanisms, I am hoping that the MIT Physics Department will implement an annual review for grad students. We already have a formal annual review for postdocs (see Sec. 4.3.4), which I find to be very valuable for solidifying the advising relationship. Sometimes a formal process can be helpful to discuss things that don't regularly come up in informal interactions.

3.3 So You Want To... Get a Roadmap for Grad School?

There are as many ways to be a successful physicist as there are successful physicists. That said, there are skills that are highly correlated with research success. If you are motivated by checking off boxes, this section list some tasks that most (though not all) graduate students in my group complete, roughly organized by when in your degree you should start. This does not mean that once you have checked a box there is nothing left for you to do or learn. Rather, many of these are lifelong activities, and starting them at the appropriate time means you are on the right track.

3.3.1 Survive Your First Years as a PhD Student?

Depending on where you did your undergraduate studies, the transition to being a PhD student at MIT can be jarring. You might have done five classes as an undergraduate and aced them all, but you now find that 2 or 3 graduate classes at MIT already saturate all of your time. Learning at MIT is sometimes likened to drinking from a firehose. On the other hand, graduate students educated at MIT are often much better prepared than students coming from other universities.

Here is a recommended checklist of things that you should do during years 1–2 as a PhD student.

- **Ask for advice.** Starting now, and for the rest of your life, you should get in the habit of asking people for advice. People are very happy to talk about their own experiences, and you will learn to ignore suggestions that are irrelevant to your situation. But if $\mathcal{O}(10)$ people tell you the same thing, then it is probably the right thing to do. (Be wary of "**survivorship bias**", though, if you only get advice from faculty members.) See Sec. A.1 for the general advice I give people at all career stages.
- **Pass the four required written exams.** You can't be a successful theoretical physicist without a deep understanding of quantum mechanics, electromagnetism, statistical mechanics, and classical mechanics. Take the written exam in these subjects as soon as you arrive at MIT, and if you don't pass on the first attempt, study up for the next attempt or take the equivalent course.
- **Immerse yourself in quantum field theory (QFT).** MIT offers three semesters of QFT, but that still isn't enough to understand the beauty and depth of this topic. QFT is the unifying language for almost all theoretical physics these days, so learn as much about it as you can. You know you've immersed yourself sufficiently if you have a favorite QFT textbook (or you hate them all and start writing your own). If offered, I strongly encourage you to take my course on supersymmetry (8.831, inherited from supergravity legend Dan Freedman); whether or not supersymmetry is realized in nature, supersymmetric QFT is a great way to better understand QFT in general.
- **Start taking your specialty and breadth classes.** There is no rush to finish the course requirements, so you might want to wait to take your specialty and breadth classes for when there is a particularly interesting offering (e.g. 8.831, wink wink). The breadth classes are intended to broaden your horizons and often lead to new research ideas.
- **Regularly attend research seminars and colloquia.** You might still be choosing your research area, and going to talks is a great way to be exposed to many different areas. Don't worry if you only understand 5% of the talks at first. Learning by osmosis/curiosity is a great way to start.
- **Read the arXiv on a semi-regular basis.** No one really reads journals any more in high-energy physics, and a paper starts accruing citations the moment it hits the arXiv. I skim the new submissions in hep-ph and hep-ex

every day. (I used to also skim hep-th, but the papers of interest to me usually get cross-listed to hep-ph.) A fun group exercise is to guess the paper content just from the title/authors/abstracts.

- **Become familiar with the Review of Particle Physics from the Particle Data Group (PDG).** The PDG is an incredible resource, and it is the first place I go if I have a particle physics question. They'll mail you a free copy (size of an old telephone book), but the online version has the same content and is much easier to carry.
- **Learn to write in LaTeX.** All reputable journals in high-energy physics use LaTeX for typesetting. Indeed, one way to identify a predatory journal is if they accept Word documents. See Sec. D.1 for more paper-writing advice.
- **Learn to prepare slides.** Most talks these days are given using slides, but it takes a lot of practice to make nice slides efficiently. Any format is fine; I use Keynote though other people swear by Google Slides or Beamer. See Sec. D.2 for more presentation advice.
- **Give a blackboard presentation of any kind.** The stereotype is true; most theoretical physicists love blackboards (or whiteboards). Even in the era of slides and screen sharing, blackboards skills are still quite important. Indeed, there are some venues (like the MIT Physics oral qualifying exam and Pappalardo Fellowship competition) that forbid slides. One nice way to gain blackboard skills is to present an article at a journal club, ideally in a field different from your own.
- **Gain public speaking experience.** Learning how speak confidently and conquer stage fright is an essential skill that requires practice. Whether you are giving a research presentation, teaching a class, communicating with the public, or emceeing an event, your "performance" is often as important as your content. You also need to learn how to think on your feet and answer questions from the audience.
- **Explain your research to non-physicists.** Your family will have no idea what you are doing in grad school, but you should try to explain what you are doing to them. This comes in handy when you start talking to reporters. (Try to do a better job than I did.)

3.3.2 Thrive During Your Middle Years as a PhD Student?

By this point, you have settled into grad student life. Now is the time to start taking more ownership of your research program. The goal at this phase is to prepare yourself to direct your own research program, and start making the transition from a student to a researcher.

Here is a recommended checklist of things that you should do during years 2–4 as a PhD student.

- **Pass your oral qualifying exam (a.k.a. Part 3).** This is the key transition point along your path to becoming a physics researcher. The process of studying is more important than the actual exam, so you might delay (or accelerate) your first part 3 attempt to make sure you have a large study group to work with.
- **Finish taking your specialty and breadth classes.** While it is tempting to leave a few PhD requirements to the end when you'll "have more time", you'll never have more time than you do now. The last years of a PhD typically involve lots of travel to present your research at conferences, workshops, and seminars, and it is hard to keep up with your courses if you are also going around giving talks.
- **Ask questions at research seminars/colloquia.** Eventually, you will be giving talks of your own, so you should learn what it feels like to ask a question so you will be better prepared to answer questions down the line. Even if

you don't actually ask your questions, preparing 1–3 questions for the Q&A period is good practice, and can help you put someone's talk in context. Wonder why I always seem to have at least one question at the end of every talk? It's because I follow my own advice.

- **Write your first (of hopefully many) research paper.** Getting your name on a paper is relatively easy if you have senior collaborators, as long as you participated actively in the research. But you really want to try to lead the paper development as much as you can, and this can start already with your first research paper. Some things you should strive to do related to paper writing (not necessarily all in the same paper, see Sec. D.1 for more details):
  - **Do the majority of calculations in a paper.**
  - **Make the majority of plots in a paper.**
  - **Write the majority of text in a paper.**
  - **Submit a paper to the arXiv.** You get bonus points if you are top of the list.
  - **Submit a paper for peer review in a journal.** Typical journals for high-energy physics are JHEP and PRD.
- **Present your research at your home institution.** Before going "on tour", get practice giving talks at MIT. You can present preliminary results, and often the process of trying to present your material can help you figure out how to structure your eventual paper.
- **Give a talk at an international conference or workshop.** Your first conference/workshop is when you declare yourself as a member of the research community. You'll want to have an awesome result to show, and you'll want to meet as many of your peers (and elders) as you can. Funding is available for these (from me or from other sources), so in most cases, you won't have to pay out-of-pocket for these (unless you exceed your travel budget).
- **Visit another institution to give a seminar.** Once you start presenting your research away from MIT, you will start to gain recognition in the field. It is also when people will start making opinions about you and your work, so you need to take these seminars very seriously. Reminder, your future employer may be in the audience. In almost all cases, the institution you visit will pay for your travel.
- **Attend a physics summer school.** One of my regrets is not going to at least one summer school during my time as a PhD (though since then, I've taught at a number of them and organized one). These are typically 1–3 week events in a beautiful place, and it is a way to meet your peers at other universities. The iconic summer school is the Theoretical Advanced Study Institute (TASI) at University of Colorado, Boulder, but there are many others. Perhaps counter-intuitively, it is best to go to one of these schools after your first research project is finished, since then you'll have something to present.
- **Form your thesis committee.** In many ways, having a thesis committee is just a formality. But I believe in having robust processes, and your thesis committee will help keep you and your advisor accountable. While you should definitely highlight key research results at thesis committee meetings, these are venues primarily to talk about your progress towards a PhD and your plans for after MIT.
- **Increase your online presence.** These days, most academics have a personal webpage, LinkedIn profile, YouTube channel, and/or Twitter account. (Links above are to my pages.) If someone hears you give a talk and wants to learn more, there should be someplace on the web that highlights your research. An online presence isn't absolutely necessary, but it can give you an edge, especially if you aren't yet well known in the field. Ideally, there

should be somewhere online where you have a more-or-less up-to-date cirriculum vitae (CV, see Sec. C.1.2), since you never know who might need to have access to that information to advocate on your behalf.

- **Start thinking about post-PhD possibilities.** Once you've experienced one full research cycle of idea to publication, you are in a position to start thinking about potential jobs after your PhD. See Sec. 3.4 for further advice about choosing your post-PhD path.

During the middle parts of your PhD, your relationship with me will start to change. As you move from the category of mentee to collaborator (and perhaps eventually to faculty colleague), you will need to tell me what you need in order to get things done, rather than the other way around. Once you start needing to boss me around to realize your goals, then you know that you are making the transition.

3.3.3 Prepare for Your Final Years as a PhD Student?

I got my PhD in 4 years. In retrospect, that was too short, and I wish I had stayed an extra year to gain more confidence and independence. But all (good) things must come to an end, so you have to leave MIT at some point. Now is the time to really decide on your future and your post-PhD path.

Here is a recommended checklist of things that you should do during years 3–6 as a PhD student.

- **Write a paper in collaboration with someone other than me.** Now is the time to grow your collaborator network and think about problems different from the ones I work on. Plus, most academic positions require three letters of recommendation, and you'll get the strongest letters from people who have direct experience collaborating with you. This can be someone from MIT or from another institution.
- **Be a peer reviewer for a journal article.** One of the signs that you are an expert in your field is that you can fairly evaluate papers in your field. Doing peer review is great practice (and a great way to see what things look like from the other side of the reviewing process). There might even be cases where I need your help with peer review, so I might ask you to co-sign a review with me (or take it over completely).
- **Agree on a graduate timeline with your advisor and thesis committee.** At least a year before you defend your PhD, your advisor and your thesis committee should know that you want to graduate. If I don't know you want to graduate, then I won't know to nominate you for various positions that require a 1 year lead time. If I don't know you want to stay another year, then I won't know to make sure I can secure funding for you. The default PhD length in the MIT CTP is 5 years; years 6 and beyond are typically supported through a teaching assistantship.
- **Apply to jobs/postdocs.** Even if you don't plan to graduate until May of $20XY$, you already have to think seriously about applying to jobs in May of $20XY - 1$. I'm not kidding. While some jobs will hire you right away, other ones have a lengthy evaluation process. The PhD thesis can wait; the job hunt cannot. In fact, there are special cases where people have started a postdoc job before having their official PhD in hand! See Sec. 3.4 and Sec. C.1 for more details, especially on the important task of getting letters of recommendation.
- **Write your PhD thesis.** If you've done things right, then this is largely a formality. You staple together your best papers, sandwich them between a nice introduction and conclusion, wrestle with formatting for a week, and you are done. Okay, maybe it takes a month (or three months if you want to write a really pedagogical introduction).
- **Defend your PhD thesis.** Again, if you've done things right, then this is just a formality. Yes, your thesis

committee will grill you. Yes, it takes time to make a nice thesis presentation. But mostly this is a chance to show off to your friends and family, and maybe get them to understand a tiny bit about what you did. As your PhD advisor, it is my job to provide cake and champagne; let me know if you have a different preferred celebratory beverage.

Even after you graduate, you will always be "my former student". Your success reflects well on me, and I will always cheer you on, no matter what way you take your career. Our relationship does not end the day you graduate, but continues as you move through, around, or outside academia. Whenever you have questions, I'll be there. Whenever you need a letter of recommendation, I'll write one. But our relationship will necessarily change, especially if you continue on in academia, since you will need to demonstrate your independence from me to get to the next level.

3.3.4 Contemplate Year Seven as a PhD Student?

The default PhD length in the MIT CTP is five years (which is being increasingly enforced as a maximum). A sixth year is sometimes necessary if you had to switch advisors at some point or if you hit a stumbling block during your quals or your research. A sixth year can also help if you get unlucky in the job/postdoc hunt. But staying for a seventh year requires a serious conversation.

When I was a grad student, the late Ann Nelson (an exceptionally creative theorist and a passionate advocate for diversity in physics) told me that all you needed was two years of research after QFT to land a postdoc. At the time, I thought she was crazy, but her perspective makes more sense to me now. If you really understand the foundations of your field, then you can have an impact in theoretical physics research in 2–3 years. You probably still need another 1–2 years to gain enough maturity and independence to thrive as a postdoc. Still, that puts most people at a 5–6 year PhD, even if they have no exposure to QFT prior to grad school. Of course, grad school is not a race, and you need to figure out the timing that makes sense to you. But whenever I feel daunted by a new project, I remember Ann's two-year rule for the time between education and impact.

3.3.5 Pass your Oral Qualifying Exam? (♯)

See Sec. G.2.1.

3.3.6 Leverage Your Thesis Committee?

At MIT, you will form your thesis committee the semester after you pass your oral qualifying exam. At some level, having a thesis committee is just a formality, since the only official thing they need to do is sign off on your PhD thesis and give you a "pass" on your PhD defense. Nevertheless, you can use your thesis committee to your advantage by meeting with them around once per year before you graduate to make sure you are on track. If you hear anyone (junior or senior) say that thesis committees are a "waste of time," I'd be happy to share anecdotes of students who would have saved themselves immerse stress if they had established their committee before facing a challenging situation.

There are three things that you should plan to discuss at each thesis meeting.

- **The progress you've already made, and plan to make, towards your PhD.** Typically, you'll need at least three peer-reviewed publications to graduate. So you definitely want to highlight the papers that have already been published, the ones in the works, and the projects you plan to initiate in the future. In addition, you should

mention talks that you've given, service work you've done for the physics community, and any relevant professional development activities. Basically, you should chart a path to graduation and show the committee that you are on track. (Or if you aren't on track, explain what adjustments you plan to make.)

- **Your plans for after MIT.** It is okay if you are undecided about what you plan to do with your PhD, but you should show that you are aware of what is needed to keep those career paths open. This is especially true if you want to apply to physics postdoc positions, which tend to have strict application deadlines, so it is important to make sure that your anticipated research progress is aligned with the academic hiring schedule. If you are thinking about an industry career, you should say whether or not you plan to do any internships before you graduate.
- **Themes and highlights from your research.** One of the reasons why thesis committees are fun for faculty is that we get to learn about awesome science, so definitely share some of the recent highlights from your research. Also use this opportunity to start developing the overarching research themes for your thesis, which can help clarify which directions you should prioritize before graduation. But try not to overstuff your thesis committee presentation with research details, since you want to leave time for questions and discussion.

Ideally, thesis committee meetings should be **around 30 minutes long**, though it is typical to schedule an hour, just in case there are important issues to discuss.

When I serve on thesis committees, here are some typical questions that I ask the candidate, which you might want to address directly in your thesis committee presentation.

- **When do you anticipate graduating?** Not only do I want to see that you are making progress towards your PhD, but I want to know your specific target graduation date (or at least graduation semester/month). Having this information is relevant both for my own planning purposes (you can't have a thesis defense without the thesis committee present!), but also to ensure that your advisor agrees that this target date is feasible.
- **If you anticipate going on the academic path, who might you ask for letters of recommendation?** It is never too early to start thinking about your future applications, and most postdoc positions require at least three letters of recommendation. In order to get three strong letters, it is worthwhile thinking about whether you might need to cultivate new research collaborations outside of your immediate research group. Indeed, one of the members of your thesis committee might be a good letter writer!
- **Is there anything the thesis committee can do to help?** Your thesis committee wants you to succeed, so if there is a particular roadblock to your PhD, go ahead and mention it. Maybe you need funds for travel, or are having trouble communicating with a collaborator, or need advice on which research direction to prioritize. Whatever it is, the thesis committee is there to help, and the yearly thesis committee meeting is a good venue to ask for support.

You might find yourself in a situation where you don't feel comfortable raising a concern with your advisor in the room. In that case, it is perfectly fine to set up a separate meeting with one of your thesis committee members to discuss privately.

3.3.7 Anticipate Possible Stumbling Blocks?

If you stay in regular communication with me, then the chances that your PhD will be derailed are relatively small. But there are some common stumbling blocks that can add unnecessary time and stress to your PhD.

- **Having a wait-and-see attitude.** If something isn't working, it likely isn't going to fix itself. Even if you have to wait for an eventual solution, articulating the problem can at least make me (and you) aware of it.
- **Relying on other people to do the work.** Physics is a collaborative enterprise, and we benefit from bringing people together with diverse expertise. When you are a PhD student, though, you should aim to understand (almost) every aspect of (almost) every project you work on. You don't want to rely so heavily on a collaborator that your project will completely halt if they stop working on it. When you are learning how to be a physics researcher, "duplicated effort" is another name for "essential cross check", so put in the extra effort.
- **Trying to find the optimal path rather than pursuing multiple ones.** If you have two possible strategies A and B, it is tempting to try to figure out which one would be "best." But unless you have a strong reason to prefer one over the other, the strategy of "A and B" is often more efficient than ruminating over your options. In many cases, "wasted effort" is actually the way to make a serendipitous discovery.
- **Pursuing grandiose ideas to the exclusion of well-defined projects.** All of us have at some point dreamed about solving quantum gravity. And one of us might even do it! But if you only think about "big ideas" then you'll miss out on opportunities to improve your research skills by working on concrete projects.
- **Generating concrete deliverables to the exclusion of big-picture thinking.** Once you learn basic research skills, you can rapidly churn out "least publishable units". But lots of incremental papers won't necessarily lead to the kinds of breakthroughs our field needs (or you need to advance your career).
- **Not knowing the existing literature.** If you aren't familiar with past research and aren't keeping up with current developments, then you might find yourself pursuing dead ends (or thoroughly well-trod paths).
- **Anchoring yourself too closely to the literature.** Sometimes, you just need to think for yourself. Yes, you risk reinventing the wheel, but sometimes you need to develop things in your own language to make progress.
- **Not writing up your work.** If it isn't on the arXiv, then it doesn't exist. If you've invested time, energy, and ideas into something you are proud of, let the world know by writing a paper about it.
- **Writing up low quality results.** There is a short-term benefit from writing lots of papers, since it looks like you are being "productive". But in the long term, this strategy will usually backfire, since it dilutes your impact and hurts your reputation.

For many of the stumbling blocks above, the key is to maintain balance. Investing too much in one approach is great when it works out, but highly detrimental if it doesn't.

3.4 So You Want To... Get a Job After Your PhD?

3.4.1 Decide Whether an Academic Career Path is Right for You?

Many beginning graduate students dream about becoming a faculty member at a college or university. Until you are immersed in research full time as a PhD student, though, it is difficult to conceptualize what it really means to follow an academic path.

A recommended time to start thinking about whether an academic career path is right for you is around the time you publish your second paper. By that point, you'll have enough information to start weighing your post-PhD options. Here are some things to consider as you begin this process:

- **You can have a successful PhD without going on in academia.** Getting a PhD forces you to develop strong research skills, which are relevant in a wide range of positions in and outside of academia. If you enjoy research, and don't mind having someone tell you what to do research on, then there are plenty of fulfilling jobs in industry.
- **You can have a successful postdoc without going on in academia.** The postdoctoral career stage offers you a chance to build your research independence with very few requirements on your time. This can be beneficial to you even if you later decide to take a job outside of academia. The downside of a postdoc is that there is no job security, and you might not have much control about where you end up geographically. But if you can handle the uncertainty (and the lower pay compared to other jobs you could get after your PhD), then a postdoc offers unparalleled intellectual freedom.
- **You have a reasonable probability of getting a faculty job if you are flexible.** While there are relatively few faculty positions at the most competitive universities, there are a relatively large number of faculty positions overall. This is especially true if you are geographically flexible and willing to take a position outside of the US. If you are reading this document, then you have probably spent some of your academic career already at a place like MIT. This means that you have substantial advantages in the academic job market, though it also means that you might be disappointed if you can't find a faculty position at an R1 university.
- **Tenure-track academic jobs are more similar than they are different**. It might naively seem that a tenure-track job at a liberal arts college, four-year university, and national lab would be quite different. From a day to day perspective, they are different, and you might have a preference for one job compared to another. But in terms of job preparation and expectations, they turn out to be rather similar. All of them involve research, teaching, mentoring, advising, and service, just to different degrees. Tenure-track jobs are very competitive, with $O(100)$ applicants for each open position, so there isn't really an "easier" job in this category.
- **Understand the difference between tenure-track and non-tenure-track jobs**. A tenure-track position involves a four to eight year evaluation period before you are considered for a permanent (tenured) position. There are non-tenure-track positions in academia, such as research scientists, teaching staff, and adjunct professors, where there is limited job security. Note that non-tenure-track positions do not typically turn into tenure-track positions, but they may be appropriate for you depending on your circumstances. There is often a large pay gap between tenure-track and non-tenure-track positions.
- **As a faculty member, you will wear many hats.** Being a faculty member is a bit like running a company, where you occupy the entire C-suite: chief scientific officer, chief financial officer, chief executive officer, etc. You are

expected to "do it all"—research, teaching, advising, mentoring, service—even though you have limited training for most of the things you are responsible for. It is an exhilarating job that can quickly turn into an overwhelming job if you don't learn how to manage your time and attention. But you get a chance to work with talented junior people for the rest of your career, which is a large part of the fun for me.

- **As a faculty member, you will struggle with work/life balance.** Work/life balance is a challenge for any job, but my sense is that it is worse in academia. Part of this is because the demands of tenure incentivize you to always be working on something. But even after you get tenure, there is always more to do and more demands on your time, and you can get guilted into saying yes even if you don't have the bandwidth.
- **There are very few jobs that offer the flexibility and professional satisfaction of academia.** For all of its flaws, academia offers unparalleled freedom to define your own path. You are anchored to some degree by your responsibilities and by grant funding. But in general you are bounded only by your ideas and ambition, so if you see an exciting opportunity, you can grab it.

Are you excited by the opportunities presented by a faculty job? If you are, then I think you owe it to yourself to seriously consider this career path. If you aren't, then know that there are plenty of jobs outside of academia that can be just as fulfilling, just in different ways.

3.4.2 Apply for Postdocs?

Most postdoc positions begin in September, though there is usually some flexibility on the start date. There is no flexibility on the application deadlines, though. Most applications are due in October or November of the previous year, though some prestigious postdoctoral fellowships have August or September deadlines; see Sec. C.4.2. To maximize your chances of landing your preferred position, you'll want to advertize your work in the spring and summer before you apply. This implies that you have to plan a whole year-and-a-half ahead of time. The earlier you start thinking about postdocs, the more informed you will be when you put in your application; see Sec. C.1.1.

Unlike grad school applications, there is no fee to apply to postdocs. Therefore, you should plan to send out a lot of postdoc applications, including to places where you think it is unlikely that you'll be hired. A rough rule of thumb is:

- **Apply to any institution where there is at least a 5% chance you might want to go there, now or sometime in the future.**

One way to think about this is that your postdoc application announces to the world that you are on the academic job market. Hiring committees often look at postdoc applications to flag candidates for future faculty positions. In this way, a postdoc application is a good way get your name out there.

Most postdoc jobs are associated with a specific research group or faculty member. Theoretical particle physicists typically have a lot of flexibility in what they work on, unless the grant funding the position is earmarked for a specific purpose. There is no centralized list of postdoc openings, and the HEP Theory Postdoc Rumor Mill only has lists of offers, not of open positions. So for most postdoc positions, you will learn about them from Academic Jobs Online (AJO), the CTP `#job-opportunities` channel on Slack, random mailing lists, scouring the websites of theory groups, or word of mouth.

There are several prestigious postdoctoral fellowships where you are expected to choose your research direction, independent of the faculty already at an institution. These are junior-faculty-light positions, so they are highly competitive

and even just getting an interview for one of these positions is a win. See App. C.4.2 for a list of these positions.

The postdoc application itself is relatively straightforward. Most applications go through Academic Jobs Online (AJO), though some institutions use custom systems. You will typically need to provide:

- ☐ **A cover letter.** This is rarely important, but can look bad if missing. Typical cover letters have one to three paragraphs saying who you are, what you have done, and why you are applying. Don't sweat it, but also don't skip it. And don't let it go over a page.
- ☐ **An academic CV.** This is a complete summary of your education, academic positions, research experience, honors and awards, mentoring, teaching, papers, presentations, grants, service, outreach, and any other relevant information. See Sec. C.1.2 for more advice.
- ☐ **A publication list.** For reasons I don't understand, sometimes a separate publication list is required, even though this information is often already in your CV.
- ☐ **A research statement.** Depending on the application, this can be 1, 2, or 3 pages long, sometimes with references, sometimes without. You'll get good at applying different "compression settings" to your various versions. The research statement should cover the highlights of your past research and convey a strong sense of what you plan to do in the future, and why it is important/exciting. See Sec. C.1.3 for more advice.
- ☐ **Any application specific information.** Read the instructions! For example, the CTP requires you to choose two potential mentors from a drop-down menu; it is remarkable how many candidates skip that simple step. An incomplete application is the same as not applying.

That's it for the part of the application you prepare. But the most important part of your application is something you only have indirect control over:

- ☐ **At least 3 letters of recommendation.** Your PhD advisor will almost certainly write one of your "core" letters; see Sec. C.2.1 if that person is me. For the other two "core" letter writers, the more senior they are and the more they know your work, the better; see Sec. C.2.2 for more advice. Some applications have a strict limit on the number of letters, while others are flexible. A 4th or 5th letter can be helpful if they provide strong complementary (and complimentary) information. A 6th letter is usually counterproductive.

Note that some readers don't even look at the CV and research statement unless there are 3 strong letters. So if you want to increase your chances, think about who might write your letters and how they can become part of your advocacy team.

The final way to increase your chances is to make sure you and your work are visible in the spring and summer before you apply. Ways to increase your visibility include:

- **Wrapping up papers.** While it okay to list a few papers as "in progress" on your CV, you want to have as many papers on the arXiv as possible before your first application deadline.
- **Presenting your work at workshops/conferences.** You should plan on devoting a fair amount of travel time to giving talks at key venues for your subfield.
- **Giving talks at your top postdoc choices.** This isn't always possible, since seminar slots are a rare commodity. But if you (or your advisor) cultivate the right connections, you can sometimes get yourself invited to speak at key institutions.

- **Having an online presence.** If someone does a Google search for you, will they find your CV? A talk you've given? A paper you've written? A LinkedIn page? Even a little bit of self promotion can help elevate your file, especially if the information online matches the information in your application.

Note that much of this will happen a year before you defend your PhD thesis. If you already have a postdoc lined up, the PhD defense is just a formality.

3.4.3 Apply for Industry Jobs?

I am not the best person to give advice about industry jobs, so I recommend seeking out help from someone who has "been there, done that." The top industry jobs are just as competitive as the top postdoc positions, but the process is substantially different.

Some things to consider if you might want to go into industry:

- **A Physics PhD alone will not get you a job.** While some industries jump at the opportunity to hire theoretical physicists with no industry-specific training, it is not universally true. Even if you think you are highly qualified, a Physics PhD by itself will not get you in the door for a quantitative or technical position. In addition, some positions have a grueling interview process that requires separate preparation.
- **Some industries are rapidly evolving.** What is true of a field at a given moment may not be true in a few years when you want to get hired. For example, data science has experienced a substantial shift in hiring practices with the rise of artificial intelligence. The world continues to specialize, and students in the academic department closest to the given industry will have an advantage. At the same time, the world continues to be uncertain, with large oscillations in hiring trends year to year.
- **Internships are highly valued by some industries.** Some sectors, like the tech sector, rely heavily on internships as a pathway to jobs. If you already know you want to work in industry, you should aim to land one or more summer internships to position yourself for a full-time position.

While I've written the advice above for industry jobs, similar considerations hold for jobs at non-profit foundations or in government. National labs are an interesting hybrid case where there are both academic-like and industry-like positions, though most theoretical physicists go on the academic-like track.

Chapter 4

Advice for Postdocs

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4.1 So You Want To... Get Advice as a Postdoc?

When writing this chapter, I imagined handing it to a postdoc who just joined the MIT Center for Theoretical Physics (CTP) (or the NSF Institute for Artificial Intelligence and Fundamental Interactions (IAIFI)). Getting a postdoc is a major achievement and significant step along the academic career path, but you still have many skills (research and otherwise) to gain along your journey.

4.1.1 Know What You Signed Up For?

For many of you, you signed up for your first real job. Congratulations! You are now a full-time employee of a university, getting benefits and earning (at last!) a living wage. You will still be drawn to the siren song of free pizza, but you will eventually cast aside your graduate school diet of instant ramen noodles.

Yes, a postdoc is a job, but who exactly are you working for? Nominally, you are working for the principal investigator (PI) of your research group (i.e. me). In theoretical physics, though, you typically have the freedom to work on whatever research problems you find interesting, with or without your PI's involvement. So really, you are working for yourself, and soon you will find that you are the toughest boss that you've ever had.

For most of you, you signed up to be postdoc because you covet the real prize: a tenure-track faculty job (or equivalent). You signed up to build your curriculum vitae and become an expert in your subfield. Believe it or not, the postdoctoral career stage is the most precious in academia, since you have all of the freedom of a faculty position without the corresponding responsibilities. But until you land that faculty job (or equivalent), you signed up for job insecurity and self doubt. Hang in there, because you also signed up to do awesome physics research.

4.2 So You Want To... Work with Jesse? (♯)

4.2.1 Be Involved in Jesse's Research Group?

As a postdoc in the CTP, you have considerable freedom to decide what research directions you want to pursue. Unless you are in a special situation with topic-specific funding (IAIFI partially falls in that category), you can really work on whatever you want with whomever you want during your time at MIT. If you want to be involved in my research group, great, you're in! I'll put your smiling face on the "group page" of my webpage, and that's that. And you are welcome to be part of other research groups as well.

To emphasize, when you were made a postdoc offer to come to MIT, you were selected by the whole CTP faculty, not just me, even if I might have been the one to officially make you an offer. If we end up working together, that's great! Or if you prefer to work with a different CTP faculty member, that's also great! Or if you prefer to build collaborations with researchers outside of MIT, similarly great! As a postdoc, you are primarily working for yourself, so you get to decide how you want to structure your time and prioritize your activities.

Now, if you want to be more directly involved in my research group, then I expect you to regularly come to my Tuesday group meetings. I also hope that you'll cultivate scientific connections with the graduate (and undergraduate) students in my group and offer them informal guidance and mentorship. But by the time you leave MIT, you should be along the path to having your own research group, and I will strive to give you the freedom and agency to pursue the research (and non-research) activities that make the most sense for your career.

4.2.2 Have Jesse as your Official Mentor? (♯)

Every postdoc at MIT is assigned an official faculty mentor, who has to file an official mentoring plan at the beginning of your appointment and do official postdoc evaluations every year to ensure you have continued work authorization. Typically, this mentor is chosen at the time of hire, though it is straightforward to switch from one mentor to another if you find a better fit. **If you want me to be your official mentor, just ask!** With high probability I'll say yes, especially if you are already coming to my group meetings. And if you want to switch to a different mentor, feel free.

The only required duty of your faculty mentor is to perform a Postdoctoral Development Annual Review; see Sec. 4.3.4. Beyond that, you should plan to have professional development conversations with me every 1-3 months. These should be separate from targeted research meetings, with a focus on your broader goals and plans. At minimum, I'll ask you to schedule one of these meetings at the beginning of every semester/summer, though you should feel empowered to meet more often as issues arise.

4.2.3 Plan for the First Few Meetings With Jesse? (♯)

4.2.4 Know What to Expect of Jesse? (♯)

4.3 So You Want To... Get a Roadmap for Your Postdoc Years? (∄)

4.3.1 Adjust During Your First Years as a Postdoc? (∄)

4.3.2 Gain Independence During Your Middle Years as a Postdoc? (∄)

4.3.3 Establish Your Identity During Your Last Years as a Postdoc? (∄)

4.3.4 Leverage Your Postdoctoral Development Annual Review?

In order for your postdoctoral appointment to be renewed every year, you have to do a Postdoctoral Development Review, typically in May. The review starts from a two part document, with Part I filled out by you to highlight the progress you have made towards your career goals and your plans for the year ahead, and Part II filled out by me to assess your past progress and the feasibility of your future plans. We then meet to discuss these two parts together to make sure we are on the same page, sign the form together, and submit to Human Resources (HR).

Here are some suggested strategies for making the most of the annual review process.

- **Focus the review on your specific needs.** HR will more or less sign off on any review form as long as it isn't empty. So the goal here isn't to check some administrative boxes but rather to really reflect on the past year and make plans for the next one. Emphasize the specific things you need to make progress, taking advantage of the fact that the form asks relatively generic questions that can be easily adjusted to fit your situation.
- **Discuss topics that might not otherwise come up in research meetings.** At some level, the formal structure of the annual review makes it easier to talk about issues that might not arise in regular meetings with me. If something is on your mind, use the annual review as an opportunity to discuss it. Of course, you are welcome to bring up any issues to me at any time, but I've found that the requirement of filling out a form can make it easier to get into the mindset of long-term planning and strategic thinking.
- **Consider bringing up uncomfortable topics.** Being a postdoc can be stressful, and there are some topics that can be uncomfortable to discuss, like balancing personal versus professional goals. While it isn't always easy to raise these kinds of issues, the annual review gives you an opening. Consider whether you might feel better if you brought some challenging issues to the surface, either with me or with another trusted mentor.
- **Ask for frank feedback from me (and give me frank feedback).** Academics are not very good at giving feedback, either positive or negative. The annual review gives you a chance to ask me directly how things are going. This is also a good time to let me know if there is something I could do to improve our mentoring relationship.
- **Use the review form as an excuse to update your CV.** The annual review will ask you about the papers you've written and talks you've given in the past year. While I am in the habit of updating my CV regularly, you might not be at this point in your career. Since you already have to go through the effort of gathering information for this form, you might as well take advantage of it to keep your CV more current. See Sec. C.1.2 for further CV advice.

4.4 So You Want To... Get a Job After Your Postdoc?

4.4.1 Have a Checklist for Faculty Readiness?

There is no sure pathway to a faculty job. Or rather, there are many different pathways to a faculty job, and therefore well-meaning people will give you different advice about what you "need to do" to get a tenure-track job. The accepted best practice for faculty search committees is to establish a rubric for evaluating candidates, but different search committee members will place different emphasis on different parts of their rubric.

Therefore, my general advice for how to establish your readiness for the faculty market is to aspire to be **"the best version of yourself"**, whatever that means. Of course, that advice is not particularly actionable, so if you are motivated by checking boxes, below are examples of different quantitative criteria that are often discussed behind closed doors at faculty search meetings.

To prove that you are ready to move from a postdoc position to a faculty faculty position, your "paper case" is made through **research portfolio metrics**, such as:

- **Number of papers.** This is a very blunt measure of research productivity, and it is well understood that paper quantity is not always correlated with paper quality. In my experience, one theorist working alone can generate one quality paper per year, though this can fluctuate depending on the complexity of your projects. But as a rough target, the number of papers you write per year should be roughly equal to your average number of collaborators, modulo a factor of 2 up or down.
- **Number of citations.** This is a very blunt measure of research impact. Citation counts often tell you more about the size of the field rather than impact of one's work. One "hit" paper can skew these numbers, and sometimes your best work can take years to gain traction. But as a rough target, the total number of citations needed to start landing on faculty short lists in theoretical particle physics is around 700, though this can float up or down by a factor of 2 depending on your specific research direction. If that number seems daunting, keep in mind that papers continue to accrue citations over time, so your file will continue to gain strength just by waiting.
- **Alternative citation metrics.** There are various attempts to try to convert bibliographic measures into more predictive numbers of long-term research impact, though I'm not convinced that they are much better in practice. Two examples that are regularly discussed are: number of papers with more than 50 citations; and h-index (at least h papers with h citations each). Again, these are rather blunt measures that don't tell the whole story. But as a rough target to demonstrate continued research impact in theoretical particle physics, you can aim for your h-index to be around 1-3 times your years past PhD.
- **Solo author papers.** Some search committee members will place an emphasis on individual research accomplishments. To demonstrate that you can work independently, consider writing at least one solo author paper at some point before applying to faculty jobs.
- **Papers in fancy journals.** Some search committee members care whether your work has appeared in a flagship journal like *Physical Review Letters* (PRL). To demonstrate that you know how the game is played, consider writing at least one PRL at some point before applying to faculty jobs.

Beyond your individual research achievements, you can use your research activities to demonstrate your **readiness to mentor junior researchers**, through:

- **Papers where you are the most senior author.** Most search committee members will want to evaluate your potential to mentor junior researchers. To demonstrate that you can lead a research group, you should aim to write at least one paper per year where you are the most senior author working with a graduate or undergraduate student.
- **Semi-official mentoring roles.** Rarely will you be the official primary mentor for a graduate or undergraduate student. But if you take on a substantial component of mentoring for a junior scientist, beyond just writing papers together, that is something to emphasize on your CV. Just make sure that both your mentee and your own mentor acknowledge your role in mentoring.

Doing great research is not enough; you also need to gain **visibility in the community**, which can be quantified through:

- **Number of talks.** This is a very blunt measure of research visibility. Typically, search committees do not distinguish between seminars and workshop talks. As a rough target, you want to give a minimum of 3 talks per year (e.g. one each in Fall, Spring, and Summer), though more than 12 talks in a year probably means that you are traveling too much.
- **Number of awards.** This is another blunt measure of research visibility. While it is rarely possible to nominate yourself for an award, if you see something that you think you would be qualified for, ask a trusted mentor to nominate you. That said, there are plenty of faculty who are hired with zero awards.
- **Range of mentors/collaborators.** At minimum, you need three senior people to write you letters of recommendation, but you also need to interact with people who are invested in your future success. These can be (un)official mentors, collaborators, or just people who would be interested to hire you for your next postdoc. The breadth of your network is one measure of your visibility in the community.
- **Self-serving service roles.** As a faculty member, you will be asked to do various service roles. Be very careful about taking on too many service roles as a postdoc, though, since your primary job is to build your research portfolio. That said, there are some jobs like seminar organizer or workshop organizer that are more self serving, since you get to interact with senior people in the field in a research context.

As you gain seniority as a postdoc, the expectations for what you should accomplish before your first faculty job will increase. If you are on your second (or third) postdoc, the expectation is that you have already had time to establish an independent research program. At that point, your broader contributions to the community might start to matter (though one has to be very careful to not let service work dominate your portfolio). Here are some examples of activities that demonstrate your **standing in the community**.

- **Peer review.** At some point, journals will start asking you to participate in peer review. This is generally a good thing to take on (within reason), both to demonstrate your research standing and to see what things look like from the other side of the review process.
- **Summer school or guest lectures.** Giving lectures at PhD summer schools can demonstrate your pedagogical skills and the trust the community places in you to educate the next generation. Similarly, you can show your commitment to teaching by giving guest lectures at university classes.
- **Advocacy papers.** These are also called white papers or position papers. For example, during the decadal Snowmass process for U.S. particle physics, there is an opportunity to provide input on promising research directions. These documents rarely involve original research, but they can help demonstrate your leadership skills.

- **Community-minded service roles.** Be very careful in taking on too many service roles, since they are rarely as valued as the time needed to do them well. That said, there are some roles for early-career scientists that can help advance a cause you care about while also demonstrating your leadership/organization skills.

Note that you do not have to do everything on the lists above to land a faculty job (and there are probably some things that I have forgotten to write down that could enhanced your CV). At the end of the day, you need to figure out the best use of your time to advance your overall portfolio. Mainly, this means being intentional about how you use your time and not assuming that things will "just work out" without some level of strategic thinking.

4.4.2 Establish Your Research Style?

As I like to say, there are as many ways to be a successful physicist as there are successful physicists. Indeed, the research community thrives when people with different research styles collaborate (and in some cases compete). This lesson can be hard to internalize, though, since you will see specific people in the community with their specific idiosyncrasies, and you will be tempted to mimic their paths to success (and avoid their pitfalls). But the reality is that there is a spectrum of successful ways to operate, and you have to figure out the mode of inquiry that works best for you.

Here are examples of dichotomies that have come up in my conversations with postdocs. I cannot give you advice about which is better, since both can be successful and there are intermediate operating points along these spectra:

- Being **a jack of all trades (breadth)** versus **a master of one (depth)**: The advantage of working on many topics is that you can flexibly respond to current developments in the field, but if you gain proficiency in a specific area, you can become the go-to person on that topic.
- Pursuing **topic-driven research** versus **technique-driven research**: If you align your research to topics of interest to your community, your work will always be relevant, but if you master a particular technique, you can build bridges to other disciplines.
- Targeting **specific funding opportunities** versus **broad funding appeal**: Funding entities use strategic calls to advance their own agenda, which you can decide to cash in on in the moment, or you can decide to make your research appealing no matter how the funding winds blow.
- Writing **one famous paper** versus **many solid contributions**: Everyone is one paper away from a trip to Stockholm, but you can still advance the field through multiple well-regarded research contributions.
- **Chasing a hot research topic** versus **advancing/starting your own research area**: If you work on a hot research topic now, then you know you'll get a lot of visibility in the short term, but there might be a greater long-term prize if you can establish the next (or next-to-next) hot topic.
- Having **a tight group of long-time collaborators** versus **a broad group of shorter-term ones**: People who know your work well can advocate convincingly on your behalf, but a chorus of positive opinions can also be effective.
- **Doing great work behind the scenes** versus **telling a great story loudly**: Eventually, all truly great ideas will become known, but good story telling can sometimes advance your agenda faster.

You only have a finite amount of time on this planet, so you should figure out which research style makes the most sense to you. And if/when you second guess your current approach, consider moving along one of the axes above to see if you can find a better fit.

Chapter 5

Advice for Junior Faculty (♯)

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5.1 So You Want To... Get Advice as a Junior Faculty Member? (♯)

5.1.1 Know What You Signed Up For? (♯)

5.1.2 Transition from a Student Mindset to a Faculty Mindset? (♯)

5.1.3 Subvert the System?

There are many written and unwritten rules in academia, and you will drive yourself crazy if you don't try to bend the rules at least a little bit. Here are some relatively safe things you can do even as a junior faculty member that will give yourself the illusion of control even if you still feel like a cog in the academic machine.

- **Rebel against at least one "requirement".** You are asked to do way too many things as a faculty member, and the people asking you to do it might keep imploring you, even if it isn't a good use of your time. Even if you are a goody two-shoes like me, you should pick at least one rebellion, where you'll just refuse to do something, no matter what. When I was a junior faculty member, my rebellion was refusing to go to commencement, so I would purposely find a reason to be out of town during commencement weekend. "But Jesse, you have to go to commencement to support your students when they cross the stage to get their diploma!" Really, I "have" to? I'd much rather take my students (and maybe their parents) to a celebratory dinner or drinks, which I view as more meaningful than playing dress up with academia regalia. "But Jesse, going to commencement is an easy way to check the 'educational commons' box for your tenure case!" Really, my tenure case will depend on whether I show up to an event where I do nothing but stand around and intermittently applaud? Look, I understand commencement is a rite of passage, which is important to students and their families. But given that I have my own family and other priorities, is being at commencement all that important? Answer: no. (As a tenured faculty member, I have attended commencement once.)
- **Leverage "unrestricted funds".** Submitting reimbursements is a pain, especially if alcohol is involved. (Once, I got audited by MIT because they couldn't believe that a single person could spend so much on one meal.) Instead of fighting tooth and nail to make sure that all of my reimbursements get processed, sometimes I allow myself to use my personal bank account if getting reimbursed would be too frustrating. Mentally, what I do is allocate one week of my summer salary as "unrestricted funds" (different from actual unrestricted funds you get from your MIT start up package). Of course, those funds are taxable and they dip into my take-home pay, but it just isn't worth the mental anguish to me to deal with a broken reimbursement system... as long as it doesn't go over 1/50th of my salary. Plus, this approach encourages me to actually take the full summer salary amount I can legally receive from my research grants.

5.2 So You Want To... Get a Roadmap Towards Tenure? (♯)

5.2.1 Get Tenure? (♯)

5.2.2 Suggest Letter Writers for Your Promotion Case?

At MIT, there are three major faculty promotion stages: Associate Professor Without Tenure (AWOT), Associate Professor With Tenure (AWIT), and Full Professor. Each of these promotions requires outside letters, roughly 10-12 for AWOT, 15-18 for AWIT, and 8 for full. For the AWOT and AWIT promotions, there is a committee of four faculty (one from each Division of the Physics Department) who prepares a list of letter writers for approval by the Department Head. One of the committee members is your "advocate", who is closest to your scientific area and usually in your Division.

At some point in the process, you will be asked for a list of 8 suggested/discouraged letter writers. Your promotion committee doesn't have to follow your recommendations, but your suggested/discouraged designations will become part of your promotion file. Here are some suggestions for building your list of 8.

- **Do the committee's job for them.** You know your research program and the people in your research field better than anyone else. So you should first try to come up with a complete list of letter writers and alternates, beyond just the official 8 you need to provide. While formally, you are only supposed to submit the 8 to the committee, your advocate will undoubtedly appreciate getting an informal longer list of potential letter writers to consider.
- **Take an "onion" approach.** You want to build a balanced slate of letter writers, from a range of backgrounds and institutions, who collectively cover all aspects of your research portfolio. Think in terms of layers of an onion, with some letter writers close to the core of your subfield, some letter writers working in adjacent areas, and some letter writers who can give a broader perspective on your work. Make sure to include some letter writers who can provide an international perspective, and consider diversity (in all its facets) in your selection.
- **Avoid "avoids".** You can recommend letter writers you want the committee to ask and ones you want the committee to avoid, with the total number capped at 8. Unless you have a huge intellectual or personal clash with someone in the field, it is best not to make any "avoid" recommendations. If you have a concern about a particular potential letter writer, bring it up with your advocate informally, and only put them in the "avoid" column if absolutely necessary. Note that the committee can still request letters from someone you've asked them to avoid, though it is marked as such in the promotion package.
- **Strategically narrow down your list.** Once you have a "complete" list of letter writers (that you have shared informally with your advocate), it is time to create your official list of 8. The goal is to maintain the overall onion balance, while choosing the people you think will write most persuasively on your behalf. You can safely omit "obvious" people like your PhD or postdoc advisors, who will be asked for letters no matter what. But you can tilt the scales in your favor by recommending letter writers who are obvious (and obviously supportive) from your perspective, but might not be obvious to the committee.
- **Be bold, within reason.** Weak letters can harm a promotion case, but so can strong letters if they are written by a weak pool of letter writers. The more accomplished your letter writers, the more weight their opinion carries. Don't be afraid of recommending the "thought leaders" in your field, though make sure to recommend people who are familiar with you and your work.

Ultimately, it is your promotion committee's responsibility to do their job and identify a strong pool of letter writers.

In many ways, the AWOT case is a dry run for the AWIT case, and sometimes letter writers get shuffled around between AWOT and AWIT for your strategic advantage. While one weak letter won't sink your case, two weak letters might, so any help you can give the committee in identifying strong letter writers will be welcome. This help is particularly welcome for the outermost layer of the onion, where you know who knows your work better than the committee does.

5.3 So You Want To... Hire Group Members? (♯)

5.3.1 Review Applications?

At some point in your professional career, you will likely have to evaluate applications for a job. I can't say much about jobs outside of academic, but within academia, there is an annual ritual (starting in mid-Fall) of reading files for many different career stages. Coming up with a way to translate an application into a score or ranking is incredibly difficult, since candidates often have complementary strengths. While I can't give you an algorithm for how to choose the "best" candidates (what does "best" even mean?), I can tell you the things I am looking for when evaluating an application, roughly ordered by priority:

- **Research experience, accomplishments, and potential.** This is the most important part of an application for an academic job, no matter what position you are applying for. The specific research is less important to me than a demonstration of research ability and strong indications of future potential. I want to see that the applicant has tackled substantial questions and has made substantial progress (calibrated of course to one's career stage). I value sustained contributions, usually with an emphasis towards recent work.
- **Research impact and vision.** The number of preprints/publications/citations is a rough measure of impact, which often goes under the banner of "productivity". I am less impressed by a long publication list, though, and more impressed if the research is making an impact on the field (or aspiring to make an impact) and/or is pursuing innovative new directions. The ideal applicant has a coherent research vision guiding their work, now and for the future, and evidence of intellectual leadership.
- **Creativity, originality, knowledge, and independence.** These are examples of "internal" research skills. All of us are one paper away from a trip to Stockholm. There are some types of progress that require grinding out a result, so I do look for evidence of perseverance and fortitude. But the most impactful work often comes from a burst of insight. I want to see that an applicant can think about problems in new ways, has a deep understanding about their field, and has the independence to pursue their own research program. I also value breadth (and depth) of technical skills and interests.
- **Ability to work collaboratively and build community.** These can be viewed as "external" research skills. Solo author papers are great, and you should consider trying to write at least one in your academic career. But most theoretical physics happens in small collaborations. I'm looking for an applicant who has made their own unique contribution in a collaborative setting, ideally with multiple different groups of collaborators. For more senior applicants, I want to see that they have helped build a community around their research field, for example by organizing workshops.
- **Written and verbal communication skills.** Doing research and communicating research are related but different skills. The ability to write, speak, and present clearly and persuasively is correlated with one's research impact, since communication is the way research results are disseminated.
- **Teaching, advising, and mentoring potential.** In addition to doing great research, an ideal applicant should show the potential to educate the next generation of theoretical physicists. This can happen through classroom teaching, group advising, or individual mentorship. For more senior applicants, I am looking at whether they have a research program that can attract and involve students.

- **Conscientiousness, collegiality, and service/leadership potential.** The academic environment is only as strong as the people in it and the connections between them. While I would like nothing more than to shirk my service duties and focus solely on research, there are multiple service roles in academia that require a research background. The ability to navigate the academic bureaucracy can be learned, but it requires a commitment to service. Ideally, an applicant is already showing nascent leadership skills.
- **Intangibles.** Does the applicant "get" what the position is about? Does their file scream "hire me!"? One has to be careful not to lean on stereotypes about what the "right" person for the job is, but often the ideal applicant has that special ineffable spark.

Reading this above list might make it seem like I am looking for an applicant with super-human skills. In many ways, I am, or at least I'm looking for an applicant who is stronger than I was at the equivalent career stage. For more context, I wrote this list to help me solidify a rubric for evaluating candidates for a junior faculty position in the CTP. So at the forefront of my mind is the challenge/opportunity of hiring a colleague I'll be engaging with for the next 25+ years.

One of the joys of academia is the opportunity to interact with incredibly talented people with diverse perspectives, so I want to hire people who can contribute to this rich community. One of the challenges of academia, though, is that you can feel like you don't belong or aren't good enough. If you are reading this document, then rest assured that you are making proactive steps to becoming a stronger applicant for your next job, in academia or elsewhere. So congratulate yourself, and make sure to read Sec. A.2.1 to help confront the inevitable feelings of imposter syndrome.

5.3.2 Evaluate Interviews? (♯)

5.4 So You Want To... Fund Your Research Program? (♯)

5.4.1 Negotiate a Startup Package? (♯)

5.4.2 Write a Grant Proposal?

Grant proposals are similar in spirit to long research statements (see App. C.1.3), but they tend to have a more rigid structure. Also, you have to make a clear distinction between past research and proposed research. The number one rule of writing grant proposals is:

- **Make your proposal easy for a peer reviewer to review.**

Your proposal should use direct language, bullet points, descriptive subject headings, topic sentences, etc. There should be no surprises in a grant proposal, and ideally, the reviewer would just have to read your proposal once to know exactly how to review it. As a mental model, think about your reviewer as needing to evaluate eight proposals over a weekend (and preparing a presentation about four of them), since that's the rough structure for DOE comparative review panels.

Here are some other guidelines to follow:

- **Lead with your most compelling stuff**. Imagine your reviewer only reads the first sentence of each paragraph (or the first paragraph of each section). Would they want to fund you on the basis of just that information? While it is a good idea to have some pedagogical/background information in your proposal, you should focus on you and your proposed research. In practice, this means front-loading your paragraphs with information specific to your work.
- **Be absolutely consistent in your story.** If you work on three things, then there should be bullet points for each thing and sections/subsections for each thing. Don't say "I work on A, B, C" and then say "I have research projects on D, E, F." Or if you do, make sure it is crystal clear to the reviewer why you are structuring your proposal the way that you are structuring it. Again, you are trying to make your proposal easy to review, and repetition is one way to help facilitate that.
- **Be absolutely clear about what you plan to do.** It is frustrating to evaluate a proposal that mixes future goals with past accomplishments. Of course, your future goals will likely be tied to your past work in some way, but don't make your reviewer have to guess what concrete projects you plan to initiate. Consider even going so far as to put past work and proposed work in different subsections so there is no confusion.
- **Ensure that your proposal mirrors the review criteria.** First, make sure you have read the review criteria, since knowing how your proposal will be evaluated will help you make a better case for yourself. Then, try to make sure you have sentences that your reviewer can more or less copy/paste into their report. If a review criteria is "What is the relation of this work to the mission of the U.S. Department of Energy (DOE)?" then there should be a sentence in your proposal that says "This research relates to [whatever aspect of] the DOE mission."
- **Highlight your previous accomplishments.** For almost all grant proposals, you have to describe previous research funded by that agency. But you certainly aren't limited to only mentioning those works, and it can be helpful to highlight your overall research track record. Most likely, your reviewer isn't going to spend time reviewing your Inspire record, but even just a few sentences about your "best hits" can help put your proposal in context.
- **Consider strategic bold face for emphasis.** A research proposal where every other sentence is in bold face is exhausting to read. But so is a proposal that looks like one monotonous blob of unstructured text. As long as you

do it sparingly, a few bold faced sentences (or key bullets) can help the reviewer focus on the most important points.

- **Consider adding targeted figures.** Some reviewers (including me!) are visual thinkers, and they will benefit from targeted figures to explain the scope of your proposed work. Don't use figures just for decoration, but a few targeted figures can be beneficial for getting your message across, including for highlighting past accomplishments and preliminary results. Because of page limits, though, figures are often the first thing that get dropped. So if a figure is important for your story, make sure to budget the space it will take up.

Grant proposals are a rare instance where you get concrete, often actionable feedback on your overall research program. So even if you don't get funded the first time around, you'll learn valuable things that will help you improve in the future. Ultimately, the goal is for your grant proposals to naturally mirror your evolving research program, such that you aren't just chasing money, but really going after the best science.

Finally, one of the best ways to improve your grant writing skills is to serve on a grant review panel or to participate in mail-in reviews. After all, you are one of the "peers" involved in peer review. While it is a non-negligible amount of work to review grant proposals (see Sec. 5.5.2), you will learn a lot about is and is not compelling in the process.

5.5 So You Want To... Perform Peer Review? (∄)

5.5.1 Review Manuscripts for a Journal? (∄)

5.5.2 Review Grant Proposals? (∄)

5.5.3 Serve on a Grant Review Panel? (∄)

5.6 So You Want To... Consider Alternative Career Opportunities? (♯)

5.6.1 Switch Faculty Jobs?

As they say, *all's fair in love and academia*. Different institutions have different strengths, and family situations can change. While you should feel a sense of responsibility to your current institution, you don't need to be absolutely loyal. If an exciting job opening comes up, especially if there is a high probability that you would prefer to switch jobs, then seize that opportunity. But of course, you'll have to navigate complex relationships and conflicts of interest as you do so.

I've never switched jobs myself, though I did once receive an informal offer from another top institution as a junior faculty member (and I receive at least one headhunting inquiry a year as a senior faculty member). So while I can't offer authoritative advice, here are some suggestions from my limited experience.

- **Act in good faith**. Don't apply to a faculty job just to get leverage at your current job. There needs to be a high probability that you would actually want to leave (a good rule of thumb is greater than 5–10%).
- **Expect your secret to get out.** Faculty transitions are delicate, and most people will try to help you keep things under wraps. But especially because you'll probably need to get reference letters, someone at your current institution will inevitably find out. So don't do/say anything that you would be embarrassed for your current colleagues to hear through the grapevine.
- **Consider being transparent.** Especially in cases where there are family reasons for your move, you might consider being transparent to your current colleagues about your situation. But this can be very delicate, especially if there are volatile personalities involved.
- **Think about where you really want to be.** Inertia is a powerful force, but once you start thinking about moving, you should take the opportunity to think about where you would be happiest, even if those places don't have current openings. That can help normalize your interest/enthusiasm level.
- **Think about what you could do to improve your current situation.** If there is something at another institution that is preferable to you, you might be able to recreate part of that where you are currently. Department chairs are often willing to spend a little money now to avoid having to spend a lot of money later to retain you if you get an outside offer.
- **Know your time horizon.** Things can change, and do change, on the 5–10 year timescale. So as you weigh your options, try to time translate out a few years to project what things might look like in the future.

Again, I have limited personal experience with the faculty shuffle, so this is definitely one to seek a second opinion about. The people I've known who have made a switch are usually much happier after their move. And the people I've known who have decided to stay have usually recommitted themselves to their current institution.

5.7 So You Want To... Manage Your Research Group? (♯)

5.7.1 Set Expectations for Your PhD Students? (♯)

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A.1 So You Want To... Get General Advice?

A.1.1 Get Jesse's Top Three Suggestions for an Aspiring Theoretical Physicist?

For me, taking a course in quantum mechanics from Antal Jevicki at Brown University was the key turning point when I realized that I wanted to pursue theoretical physics as a career. But it was not until I went to graduate school that I realized exactly what it means to be on the front lines of scientific progress. Until you experience the simultaneous frustration and exhilaration of research, it is hard to really grasp what it means to be theoretical physicist on a day-to-day basis.

When I taught quantum mechanics myself at MIT, I gave the following advice to my students on the last day of class (mostly juniors, many of whom would go on to graduate school in physics and related fields). I've since repeated versions of this advice many times over the years, to people at a variety of career stages., so I'll repeat myself again here:

- **Find mentors**. Even now as a tenured faculty member, I have around five senior colleagues I regularly turn to for advice. As a younger scientist, it is even more important to have someone (and preferably multiple people) who are looking out for your best interests and giving you honest feedback. In most cases, your mentor will also be your research advisor, but it is generally a good idea to also have a mentor outside of your research group as well. Your mentors will often be your strongest advocates when it comes time to get a permanent job in academia (or elsewhere).
- **Be visible**. Somehow society's image of a theoretical physicist is a lone genius toiling away in a closed office. The reality is that physics (especially theoretical physics) is a social enterprise, with many research ideas arising at the coffee machines (or at the lift lines if you go to winter conferences). As a younger scientist, you might feel that the best thing you can do is focus on your specific research project and exclude the outside world, but in my experience, making yourself visible is a better way to forward your research career. This means that you should go to as many seminars and colloquia as you can (and even try to go to dinner with the speaker), as well as discuss your work regularly with people outside of your immediate research group. For me, I credit regular lunches with my theory colleagues with saving me from the research doldrums when I was a postdoc at UC Berkeley.
- **Tell a story**. Science is a process of discovering the ultimate truths of nature, and while the truths themselves are independent of the research process, research itself is shaped by the personalities involved. In order to make sure that other scientists understand and appreciate your work, you need to make an effort to explain not just the results of your work, but why your work is interesting and important and how it fits into the narrative arc of the field. This means that you have to develop strong writing and presentation skills. Bland research results are difficult to appreciate, but telling a compelling story about your research (in print or in person) is a great way to engage your audience.

I often add "**Get a second opinion**" as a fourth tip. I certainly don't have a monopoly on good advice, and by synthesizing the perspective of multiple people, you can better figure out what works for you and your situation. In that spirit...

A.1.2 Get a Second Opinion?

If you are looking for more advice, you can see what my current and former group members have to say in Sec. B. Some other potentially helpful links:

- **Peter Fisher, "Fisher Files".** This is a transcript of a podcast by my MIT colleague Peter Fisher, which I found to be quite enlightening (especially the part about the fish).
- **Gerard 't Hooft, "How to become a GOOD Theoretical Physicist".** This is a bibliography of topics for aspiring theoretical physicists. It is certainly not complete, but a starting point to find information on general topics. He also has advice on becoming a bad theoretical physicist.
- **Steven Weinberg, "Four Golden Lessons".** One has to be careful taking advice from a Nobel laureate, but this is a classic article from one of the giants of theoretical physics.
- **Jeffrey Linsky, "Unexpected Advice for Beginning Graduate Students in Astrophysics".** When I taught 8.398 (the require graduate seminar for first-year Physics PhD students), I often assigned this to the students to read. The format of it inspired some of this document.
- **Martin A. Schwartz, "The importance of stupidity in scientific research".** A nice reminder about how scientific research actually works.
- **Chris Quigg, "Advice to a lecturer".** Advice specifically about giving talks.

Do you have other recommended links? Let me know!

A.1.3 Hear a "Theory of Everything" for Advice?

I remember going to a biophysics talk many years ago, where the speaker claimed that physicists should be jealous of biologists, since biologists had a grand unifying principle for their field, namely natural selection. I don't know if the speaker was joking (especially since other effects like genetic drift are known to be important in biology). But there is something satisfying about having a concise "theory of everything", even if it is oversimplified.

Do I have a grand unifying principle for advice? Not really, but there is one piece of advice that is generically helpful if you are confused about what to do in almost any situation:

- **The desired outcome determines the required input.**

To see how this advice might work in practice, let's take the example of getting tenure. At MIT, the process for getting tenure involves (among other things) a committee asking for 12–18 external letters of evaluation about the candidates research program (see Sec. 5.2.2). So for the desired outcome of tenure, the required input is a dozen-and-a-half glowing letters. Now, work backwards. If the desired outcome is a dozen-and-a-half glowing letters, what is the required input? Intuitively, you'll get stronger letters from referees who know you and your work well, so that means you'll need to visit and give talks at the dozen-and-a-half institutions where your referees are likely to be. If the desired outcome is giving a bunch of great talks, what is the required input? You'll need great research results that you organize into a compelling story. And so on. When viewed this way, the process of getting tenure seems a lot more straightforward (though no less daunting).

You can apply this logic to all sorts of things in life, and it can help figure out what is valuable and what is not. For example, grad students sometimes ask whether grades matter in grad school. Well, it does, only to the extent that someone actually looks at your transcript. If the desired outcome is getting a good job, then you have to ask whether a transcript is a required input. In most cases, the answer is no. So find out what the required inputs are for the job you seek, and that can help you prioritize your time.

A.1.4 Follow Best Practices in Physics and Beyond?

Over the years, I've collected a bunch of small pieces of advice that don't easily fit into categories. Here are a few of them, in no particular order.

- **Use shared documents.** Want to get something done? You'll often be more motivated if there is a group of people working in the same space. Indeed, SYWT started off as a shared Overleaf document with my research group, which was the motivation I needed to keep developing it.
- **Use backward design when you want to achieve specific goals.** This is similar to the advice in Sec. A.1.3. While some activities are open ended and the ultimate outcome cannot be foreseen, many activities have specific end goals. Starting from the goals and working backwards is often more effective than shoehorning existing content into the goals at the end.
- **Know your personality preferences to know your blind spots.** There is no "right" personality type to be a physicist. But you should know your preferences since that will help you see problems from multiple perspectives and more effectively collaborate with different types of people. If you are naturally extroverted, think about how an introvert might approach giving a presentation. If you prefer abstract reasoning, think about how someone who prefers concrete examples might approach a problem. There is a lot to learn from consciously shifting your mindset every once in a while.
- **Sit in the front row.** When you go to a seminar/colloquium/lecture/etc., how often do you sit in the front row and try to engage for the whole event? While there are certainly some times when you need to attend an event "just for appearances", most of the time, you are going to an event to actually learn something. Force yourself to be engaged (and off your laptop) by sitting in the front row. Not only will this enhance your own experience, but it will make the speaker feel more welcome and often spark more interactions with the audience.
- **Be a citizen of the world.** One of the benefits of academia is that you get many opportunities to travel. Take advantage of this to explore new places and expand your horizons. When possible, cultivate friendships with colleagues in places you visit often.
- **If you don't position yourself, then the world will position you**. For better or worse, the labels that people assign to you (consciously or not) will affect your trajectory. If you earn a reputation for doing X well, then people will think of you as "the person who does X well". If people's label of you doesn't match your self image, you'll be unhappy. So be proactive at positioning yourself where you want to be in the community.
- **It's worth the money if it helps your livelihood.** This one comes from my family friend Chris Davis who is a potter in Maine. When I was in high school, she mentioned buying $2000 shelves for her kiln. I thought that was an outrageous amount of money for a relatively small piece of equipment. But from Chris's perspective, it was thinner than her previous shelf (so she could pack more pots in her kiln) as well as lighter (so it was easier on her back to move it), making it was a no-brainer to buy it. So buy the nice chalk (and a nice laptop) if it improves your science output.
- **Consider whether the opposite of what you believe might be true.** Society is full of black and white thinking, whereas the reality can be more nuanced. If you have a strongly held belief, or even just a knee-jerk intuition, consider whether there might be any benefit in taking the opposite view. For me, I have had to reconsider (and re-reconsider) my opinion on standardized testing for graduate admissions, since what seemed like "obvious"

positions at the time turned out to have surprising implications.

- **Senior people in your field are the closest you'll get to a time machine.** Want to know what your future self might look like? Look to your elders! Whether you are inspired (and want to follow in their footsteps) or dismayed (and want to chart a different path), you will learn a lot from people who have "been there, done that". Most importantly, don't assume that your situation when you get older will be dramatically different, unless you make an active effort to change things. This is especially true in academia, where structural changes can be glacially slow to implement, even when there is widespread agreement that change should happen.

Do you have other gems you'd like to share? Let me know!

A.2 So You Want To... Manage Stress and Anxiety? (卦)

A.2.1 Confront Imposter Syndrome? (卦)

A.2.2 Learn Some Mental Health Techniques? (卦)

A.2.3 Recognize Unhealthy Procrastination?

All of us have procrastinated at some point in our lives, perhaps even on a semi-regular basis. (Indeed, as I'm writing this, I am procrastinating working on a long-overdue draft of a paper.) I would argue that some degree of procrastination is actually healthy, since sometimes we procrastinate when we don't have a clear sense of what we want to accomplish. In such circumstances, distracting our conscious brain can help our subconscious find a better path forward.

Procrastination becomes a problem, though, when it becomes a permanent state of being. Do you regularly miss deadlines or require constant reminders? Do you "precrastinate" on time-insensitive tasks to avoid tackling time-sensitive ones? Are you missing out on new opportunities, since you already have a backlog of previous commitments? Or maybe you take on new commitments as a way to distract yourself from other things you should be doing?

It is important to distinguish unhealthy procrastination from two healthy time management strategies: prioritization and ritual. With **prioritization**, you are acknowledging that there are only so many hours in the day. Thus, when you let the ball drop on something less important in order to take on something more important, then you aren't really procrastinating but rather setting your priorities. On the other hand, less important tasks can still be urgent tasks, and prioritization turns into procrastination when you regularly avoid doing time-sensitive tasks. For example, if there is something that will only take you an hour, but you put it off for two weeks, then you are probably engaging in unhealthy procrastination.

With **ritual**, you are acknowledging that you are more efficient when you have a set schedule. Thus, when you postpone doing an important task because you have scheduled time to do something less important, then you aren't really procrastinating but rather respecting your calendar. On the other hand, there is a perverse comfort in doing a slew of mindless tasks, and ritual turns into procrastination if you let pre-established busywork time blocks extend beyond their nominal boundaries. For example, if you let a one-hour scheduled email session balloon to fill the entire morning, then you are probably engaging in unhealthy procrastination.

When I worry that I might be procrastinating on some onerous tasks, I ask myself the following question:

- **"What are you waiting for?"**

Here are some typical answers to this question, with my followup actions:

- **"I am waiting to get some crucial piece of information."** If that's the case, then I immediately send a message to the person responsible for getting that information to me.
- **"I am waiting to get an uninterrupted block of time to work on this task."** If that's the case, then I immediate put a block in my calendar of a sufficient length to make substantial progress.
- **"Well, I am not sure if this task is even worth doing."** This is where healthy procrastination can be helpful in figuring out whether some task should simply be dropped from my to-do list. If I can't decide in the moment, I add an item to my to-do list to evaluate whether the task is important enough for me to do or not.

- **"I'm not waiting for anything, but I just don't want to do it."** This is a clear sign that I'm engaging in unhealthy procrastination. This response usually goes hand-in-hand with my perfectionism, where I don't want to finish doing something since then I would have to "lock in" an imperfect result. Once I've recognized the problem, I put a block in my calendar to simply start working on the task, which is usually enough to get me over the hump.

Sometimes, I can play a trick on myself if I have two onerous tasks to do, where I purposely procrastinate working on one of the tasks in order to make progress on the other. This is still a kind of unhealthy procrastination, but at least something gets done rather than nothing.

Finally, depending on your career stage, one potential way to address procrastination is through delegation, where you ask someone else to do the task for you, or at least put together a first draft. But this should be used sparing, since if there is something that can be delegated, then it should be delegated as soon as possible to avoid it languishing on your to-do list.

A.2.4 Seek Professional Help?

If you are experiencing mental health challenges outside of physics, please consider seeking professional help. You cannot do your best work if there are external factors in the way. If you have any thoughts related to self harm or suicide, please seek professional help immediately. There are several outlets through which MIT can facilitate your interaction with extra resources and mental health services, including:

- **Student Mental Health & Counseling**: 617-253-2916 (days), 617-253-4481 (nights and weekends)
- **Student Support Services**: `s3-support@mit.edu`
- **GradSupport**: Room 3-138, 617-253-4860
- **DoingWell at MIT**
- **Mental Health and Wellbeing for Graduate Students**

In the spirit of trying destigmatize seeking help, let me provide an anecdote from my own experience. When I was a grad student, my younger brother Aaron was diagnosed with brain cancer. This was the first time in my life that I experienced a true panic attack and an extended depressive episode. After struggling by myself to try to resolve the complex emotions I felt, I sought out help from my primary care physician (to prescribe medication to handle the anxiety) and from a cognitive behavioral therapist (to help overcome my feeling that I had to live "two lives" to make up for my brother's illness). Without this intervention, I would have not made it through grad school. Indeed, my performance in a string theory class that semester was far worse than in any other class I had ever taken. But I was able to make it through this mental health challenge with professional help.

Later, when I was a junior faculty member, my brother passed away, and the same feelings of anxiety and depression came back, though less extreme. This time, I went to a psychotherapist, and I found it helpful to just talk through my mental health challenges. Of course, my family (and my students and my physics colleagues) were supportive through this episode, but talking to someone distanced from my personal and physics life helped me gain a broader perspective and learn more effective coping strategies.

A.3 So You Want To... Be Introspective? (ㄔ)

A.3.1 Gauge your Abilities? (ㄔ)

A.3.2 Overcome a Research Roadblock? (ㄔ)

A.4 So You Want To... Navigate Difficult Situations? (∄)

A.4.1 Handle Inappropriate Conduct Directed Towards You? (∄)

A.4.2 Handle Inappropriate Conduct Directed Towards Others? (∄)

A.4.3 Confront Someone who has Plagiarized Your Work? (∄)

A.4.4 Help a Peer in a Personal Crisis? (∄)

A.4.5 Understand the Challenges faced by Underrepresented Groups in Physics? (∄)

Appendix B

Advice from Jesse's Group Members

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B.1 So You Want To... Understand Why This Appendix Exists?

In this appendix, there is more or less verbatim advice from various members of my group, both past and present. Your mileage may vary.

The appendix exists because SYWT started as a shared Overleaf document, which my group members gave me ample feedback on and made helpful edits to. In some cases, it made sense for me to incorporate their thoughts in my own words. In other cases, though, their (lightly edited) advice was more powerful, which is why I put it here.

If you are a current or former group member of mine, and you want to contribute something in this appendix, please send it to me, and I'll likely include it mostly as is.

B.2 So You Want To... Get Advice from Samuel Alipour-fard?

- **Develop a healthy relationship with your own suffering.** You will always be suffering in one way or another, but this is not necessarily something that you need to freak out about. Developing a healthy relationship with your suffering may help you step back and understand it. Understanding your suffering may help you grow and empathize with others in similar mental states. I think this holds for any human being. However, because graduate school is notoriously correlated with anxiety, depression, and similar "negative" mental states, I think it is particularly helpful for graduate students to start developing practices around this idea early.

B.3 So You Want To... Get Advice from Sean Benevedes?

Here is Sean's advice on writing a research statement, which complements the advice in Sec. C.1.3:

- **Know what your point is.** Think about what you want this statement to convey. It is very easy to fall into the trap of writing aimlessly, putting words on the page to fill space without thinking about what you're trying to say with your statement and how your writing accomplishes this; remember that the desired outcome (convincing the people reading this of something) determines the required input (your writing), so you need to know the desired outcome. It's also totally fine if this changes during the process of writing; you might realize that your point isn't what you thought, which is great because it means you've learned something!
- **Structure your writing to communicate your point.** If you're trying to convey that you're equipped to do x research or fill y position because you have z skills, you should structure your statement to convey that as clearly as possible. This means putting this "thesis statement" in your statement early (first sentence of the introduction is great) and connecting to it often with evidence (i.e. your experiences and how they helped you cultivate your relevant skills). As an example of what this does **not** look like, I wrote a research statement where I just chronologically summarized my research projects and things I learned in each; without connecting this to my broader point, I basically just created a narrated CV! However, by adding a few statements in the introduction with careful use of boldface emphasizing the big picture, and by referencing this throughout the statement, it became much stronger.

Here is additional advice from Sean:

- **Prioritize making time for physical activity**. In addition to being generally "good for you" on several levels, having some amount of exercise on a routine basis wards off brain fog and can make you more productive overall.

B.4 So You Want To... Get Advice from Cari Cesarotti?

- **Never just do something; understand why you're doing it.**
- **Just write the freaking paper!**

B.5 So You Want To... Get Advice from Rikab Gambhir?

B.5.1 Get Rikab's Advice on Writing a Research Statement?

Here is Rikab's advice on writing a research statement, which complements the advice in Sec. C.1.3:

- **Find your theme.** The research statement should be more than just "I accomplished X. Then, I accomplished Y, before moving on to Z", even if writing this way is technically still all about you. There should be some sort of underlying thread behind the things you talk about that you keep emphasizing. Even if X and Y are completely unrelated, you can find some geodesic connecting them – "Using the experience gained working on project X, I found a new Y that I was able to apply that same experience to Z".
- **Weak Anonymity Condition.** The research statement should be about you, and *only* you – not your field, not your results, not your collaborators. Only talk about them insofar as they relate to you! Your essay should be such that if your name was removed from the essay, it should still be identifiable as yours!
- **Strong Anonymity Condition.** This is hard to achieve, but a stronger version of the W.A.C. – any sentence should be identifiable as yours. More concretely, all your sentences should contain *information*, in that each sentence should be something that only you (or a small subset of people) could have said!
- **The Pareto Principle.** The Pareto Principle tells us that 20% of the sentences should contain 80% of the information (weighted by importance of the information)! These should be frontloaded – the introduction paragraph and topic sentences.
- **The "Who Cares?" Test.** Any clause in your essay should have a positive answer to the question "Does the audience really care?" This is to say, avoid needless pedagogy or backstory – the audience cares about you and why your accomplishments are important in the context of the field, not the specifics of the accomplishments.

B.6 So You Want To... Get Advice from Lina Necib?

B.6.1 Get Lina's Advice on Building a Professional Relationship with Your Mentor?

As a former Jesse student and a current professor, I have had a long reflection on this, and as funny as it is, Jesse gave me this as homework (yes, he still gives homework even if the student is a professor now!)

I will write this more as "A few things that Jesse kept telling me and I never understood until after I left":

- Jesse, or more generally, your advisor's job is to help you! That's what they are there for. If they are asking you tough questions alone in an office, although that sounds intimidating, the advisor is trying to teach you something. (Happy to discuss how much I was sweating the first time Jesse did that to me).
- I have always felt like I had to impress my advisor. That is NOT the dynamic you're looking for. **YOUR ADVISOR IS ON YOUR TEAM!** They are there to get you to where you want to be, not to judge you and grill you for sport. (At least Jesse is definitely not).
- Use your advisor's time efficiently! Jesse told me this in grad school and I had NO idea what that meant. Now I do. Advisors have a limited amount of time, but they really want to help you. That means, optimize meeting time: bring a list of questions, confusions, practice talks. Remember, if it looks like you're pushing hard on a paper/project, Jesse will get on that train too because he will be excited about the results, and he will push for your paper to get out. Win-Win!
- Communication is key. I was extremely intimidated by Jesse in the first couple of years, so much so that I was not reaching out to him for help which I later realized I needed. Talk to your advisor about your struggles, they can help or point you to the right resource. They can also adjust expectations if you're going through a rough period. That will reduce frustration from you and your advisor, since that usually stems from expectations not being met.
- Jesse once said "I can see you have a lot of feelings, explain them to me. If you're angry, it means you want more control over the project; if you're anxious, it means you're overwhelmed; if you're sad, it means you don't find things interesting," or something along those lines. So whatever is the cause of a frustration with a project is, communicate it.
- If you're not matching in personality styles, or professional styles with your advisor, it's okay to politely tell them so, and it's perfectly okay to leave their group. You will succeed regardless (you made it here!), you just need the right person to help you do that. If your advisor is causing you stress and anxiety because you think they are a little too intense or intimidating for you, it's okay to find a different advisor.

B.7 So You Want To... Get Advice from Pedro Rivera-Cardona? (♯)

[Haven't editted yet. –jdt]

Framework for Intentional Mentoring

RRIA (Reflective, Responsive, Inclusive, Adaptive).

Mentoring is reflective of how you've mentored others and by others.

Mentoring must be responsive to your mentee's goals, needs, skills, etc.

Fluidity between advocate and supervisor. It's a spectrum!

Mentoring needs to be inclusive of the structural and interpersonal barriers to inclusion.

Equitable treatment is not the same as identical treatment.

Make implicit expectations be explicit to your mentee.

Mentoring approaches need to be adaptive to individual and systemic disruptions.

Anxiety about not knowing what to do, barriers to inclusion (stereotypes, hidden curriculum), etc. can disrupt effective mentoring.

Gather data from your positive mentorship experiences and your mentee's background and needs to create a unique one-on-one relationship.

B.8 So You Want To... Get Advice from Sokratis Trifinopoulos?

When it comes to research BE COURAGEOUS! This means two things: do not shy away from opportunities to initiate projects and indulge yourself in the exploration of our unfathomably rich field. Participating in multiple projects will enhance your research experience, facilitate future collaborations ,and eventually lead to a desirable "highly productive resume".

How to get involved? Rest assured, MIT is an environment that guarantees those opportunities. Just talk to people about what they are excited about, go to the seminar talks (see relevant bullet in Sec. 3.3), and eventually simply express directly your interest asking if there is any available task or ideally propose your own ideas. Beware though, it is important that you know your limits, because putting two many eggs in the same basket can have the reverse effect.

However, one of the most effective ways to learn those limits in the first place is testing them yourself. I found that it is preferable to put yourself in a challenging situation rather than let circumstances do that for you, so try to have a proactive approach. Then you might also discover that those limits can be expanded! For example, collaborating with bright and enthusiastic people will transform your own productivity and skillset on a daily basis.

The second part of my advice concerns the scope of your research interests. To begin with, following the advice of your professor regarding your main research topic does maximize its chances of being impactful. However, while not necessary in the beginning, scientific independence can prove very significant on your academic path later on. Towards that goal, try to keep an open eye on what else could be out there, starting from what the other groups in CTP/IAIFI are doing and even going beyond to whatever direction captivates your imagination.

Here the trap is that, working on multiple topics at the same time, might slow down your progress towards becoming an expert in one particular direction. Our community does indeed value expertise quite highly and it is what makes people easily identifiable. A simple guideline, that can prevent you from getting absorbed in too many things is: "Try to spend no more than 15% of your research time on cultivating your own new ideas (disclaimer: non-conventional ones are encouraged!)". They don't have to be ground-breaking from the very beginning, just something to get you started. Personally, I feel highly motivated to learn more about a new topic when I already have a project to work on. Besides enhancing invaluable qualities like creativity and scientific confidence, this exploration is first and foremost a lot of fun!

Appendix C

Advice about Academic Jobs

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C.1 So You Want To... Prepare Application Materials?

Depending on your career stage, the precise formatting and content of your application materials will change. But below is generic advice that is appropriate for multiple career stages. Note that I am thinking primarily about academic research jobs, and **some of the advice below may be counter-productive for industry or teaching-focused jobs**.

The most important advice I can give is:

- **Start early and get feedback often.**

You'll be surprised how much time and effort it takes to prepare an application. And you'll be grateful to the people who help you revise (and re-revise) your application materials.

C.1.1 Think About the Criteria for Getting a Job?

For almost any job that you will be applying to, you will need to demonstrate that you are already developing the skills needed to carry out that job. For example, the primary job of a PhD student is to carry out research, so when you apply for PhD programs, you'll need to demonstrate the progress you are making towards building your research skills. The job of a faculty member involves research, teaching, mentoring, advising, and service, so your faculty application/interview will need to highlight your development along those axes.

Basically, for whatever the next career step you want to pursue, you want to start developing the skills needed for that step in advance of taking that step. The emphasis here is on the word "start". No one expects you to be a fully formed PhD student when you begin a PhD program. No one expect you to understand all of the intricacies of a faculty job until you actually get that title. People evolve and grow into their roles. But if you haven't demonstrated the ability to make the next step, it will be hard to convince someone to hire you. The emphasis here is on the word "demonstrate".

There are some jobs where they assure you that no background is needed. For example, quantitative finance jobs will often say that no knowledge of finance is needed and that they will train you on the job. Sure, the specific knowledge about finance isn't needed, but the general research abilities of a PhD student/postdoc is essential for those kinds of positions. This example highlights that for many jobs, a broad skill set is much more important than specialized domain knowledge, which is why physicists work in such a broad range of disciplines after getting their degrees.

C.1.2 Write a Curriculum Vitae?

A CV (curriculum vitae) is a complete accounting of your academic career. This distinguishes it from a résumé, which is a one-to-two page summary of your qualifications. Most academic positions require a CV, not a résumé, while it is reversed for industry. You can see some example CVs here:

- Patrick Komiske (May 2021, before applying to postdocs)
- Jesse Thaler (May 2022, before being nominated for a prize)

Some institutions require CVs to be in a specific format, though that is rare.

An academic CV should contain the following information, in roughly this order:

- **Header.** The top of your CV should certainly contain your name and contact information. It typically contains the words "Cirriculum Vitae" or "CV", and the date it was last updated. Do not include any personal information like

age or marital status, unless explicitly required. (Some level of personal information is standard for European CVs, but not for North American ones.)

- **Employment.** List all of the major jobs you have had. Being a graduate student doesn't usually count, so most grad students drop this section entirely. (This is changing, however, with the increase in collective bargaining agreements.) Don't include "small" jobs unless somehow relevant to your application. Major jobs include faculty positions, postdocs, research fellowships or internships, and your work history if you had jobs outside of academia.
- **Education.** Include your graduate and undergraduate institution, in reverse chronological order. Unless you did something exceptional in high school (or are still a young undergraduate), you can omit high school experience. For each institution, provide your degree(s), years attended, and if applicable, the name of your advisor and your thesis. If you do not yet have a degree, provide the anticipated degree date, with the word "anticipated". In most cases, every position in your CV should have both a starting and an ending date.
- **Honors and Awards.** If you received honors or awards for your work, list them here in reverse chronological order, with the years attached. This can include things such as fellowships, teaching prizes, and honor societies, although awards such as "graduated with highest honors" would more properly belong in your education section.
- **Publications and Preprints.** List all publications/preprints where you are an author and made major contributions, in reverse chronological order. I also like to number the papers in reverse, so that the first paper number is a large number, instead of "[1]". You can optionally add a 1 sentence blurb to each paper describing the project or your contributions. Optionally, you can add a "Works in Progress" section, to inform the reader of your current interests, but this should be used sparingly. For unknown reasons, some applications request that publications/preprints go in a separate document.
- **Talks.** A list of all talks you have given, in reverse chronological order. This includes everything from invited talks to colloquia to seminars to guest lectures. All listings should include the date, the title, the location, and the name of the event. If the talk was virtual, say so. Some people provide a link to the pdf/video or other talk resources. If you have given the same talk multiple times at different locations, you can list each separate talk once, and underneath list all the times you gave it.
- **Mentorship.** If you have mentored students, even informally, list them here. Make sure to include the capacity in which you mentored them.
- **Teaching.** A list of all classes you have taught or participated in teaching. This includes even just being a grading teaching assistant!
- **Leadership, Service, Outreach, etc.** A CV should include all relevant academic activities, which includes leadership roles, committee service, and public engagement. Sometimes it is valuable to include a list of skills (e.g. specific programming languages you know), though one typically does not list hobbies.

Depending on the specific position you are applying to, it makes sense to customize the order of your entries, such that the most important things go first. For example, when I went up for tenure at MIT, I put "Mentoring", "Teaching", and "Service" before "Publications and Preprints"; those three categories are explicitly part of MIT tenure criteria, and I didn't want them to be glossed over by someone scrolling through my publications.

I am told that the average CV only gets looked at for 1-2 minutes, so the structure/presentation really matters. Hand

your CV to a trusted friend and ask them what stands out when they quickly skim it; you'll know you need to make adjustments if they don't immediately focus on the key content. To gain even more insight, collect as many CVs as you can from people at various career stages. If you see something that works really well, copy it (format not content)! Plus, seeing the evolution of people's CVs over time can give you a better sense of the academic road map.

While the CV is supposed to be complete, it is possible to go overboard. Consider omitting or consolidating minor activities. For a webpage, you can make a short CV with just the important information and a separate long CV with everything.

C.1.3 Write a Research Statement?

Whether you are applying for a PhD program or going up for tenure, you will almost certainly have to submit a short statement about your research interests. This is sometimes called a "statement of purpose" or "research narrative". Here is some advice about this document, whatever it is called:

- **Keep it short!** In most cases, you will be aiming for 2 pages, 3 pages maximum. Some places only allow 1 page, so check the rules. You can sometimes fudge the margins / font size if you need to but don't go overboard. This page limit will help you focus your writing on the key points.
- **Read the previous bullet again!** Even the research statement for my tenure package was only 2 pages long. Unless you get explicit permission to write something longer, keep it short. Some information will need to be summarized into lists, which is fine as long as not everything turns into lists.
- **Make your research statement about <u>you</u>.** A research statement is not the place to give a pedagogical summary of your field. It is more like an advertisement where you and your accomplishments are front and center. Everything that you write should be related to you in some way, such that if your name were removed from the file, it would still be possible to figure out who you were from the content.
- **Use bullets and bold face for emphasis**. Do you work on three topics? Then consider providing those three topics in the form of bullets and/or bold-faced headers. This can help the reader identify the key information about your research program. (Hmm, interesting that this list is given as bullets with short bold-faced headers...)
- **Be consistent and use parallel structures.** Do you work on topics A, B, and C? Then consider writing a paragraph on each with the same overall logical form. Make sure any list that mentions A, B, and C is put in $A/B/C$ order, otherwise your reader will get confused.
- **Write for the weary reader.** The person reading your research statement might be reading $O(100)$ similar files. Imagine that they only read the first sentence of your research statement (and skim the rest). Would they want to hire you based on that information alone? What if they only read the first paragraph? Or first sentence of every paragraph? Or just the sentences that you've strategically put in bold face? Do your accomplishments and potential come through? Do you capture your reader's interest enough that they will want to read more carefully?
- **Make your topic sentences count.** Related to the above, front-load your writing by mentioning your research first and providing background material second. For example, instead of saying "X is important. I work on X.", you can say "I work on X. X is important because of Y." Not only is that more engaging writing, but it also makes you look more like an expert in the field. This is sometimes called "ice cream cone" writing, since you are putting the sweet stuff on top.

- **Use references strategically.** Some research statements don't have any references. But if you want to include references in yours, prioritize citing your own work, particularly papers that are high impact or explore a new direction. Make sure to check the rules about whether references count towards the page limit. If they do, you can sometimes cite papers by the number in your CV. If they don't, be selective and don't just cite everything, since the full list will already be in your CV.
- **Consider customization.** By adding a customized paragraph for different applications, you can highlight your interest in and fit to a specific program. For example, when applying to PhD programs, it is often helpful to list specific faculty members you would be interested in working with. Similarly, when applying to postdoc or faculty positions, it is often helpful to highlight the unique synergies that would amplify your research program. Of course, if you do this, be very careful to send the right customized statement to the right institution!
- **End on a high note.** One way to guarantee this is to write a short summary paragraph at the end. Get your self-promotion juices flowing!
- **Get feedback.** This might seem obvious, but you want to get feedback both from experts in your discipline as well as from non-experts who can help you with grammar and clarity. Until the point when I got tenure, my wife read all of my research statements, and her suggestions were often more valuable than those from my physics colleagues.

See Sec. B.5.1 for complementary advice from Rikab about writing research statements.

C.1.4 Write a Personal, Teaching, Mentoring, or Diversity Statement?

Much of the advice in Sec. C.1.3 is relevant for other types of statements you might be asked for in an application. This includes a "personal statement", "teaching statement", "mentoring statement", "diversity statement", etc. But there are some important differences between these documents that are useful to understand. The most important thing to keep in mind is:

- **Read the instructions.**

Some (but not all) applications will be specific about the kinds of information they want in these statements. So make sure to follow those guidelines, which overrides any of the advice I give below.

If the guidelines are ambiguous, then for most research positions, **"personal statement" is more-or-less synonymous with "research statement"**. A key exception is if the application asks for a separate "research statement" or "research proposal". In that case, the personal statement will be a more general overview of your research trajectory and future goals, whereas the research proposal will be about a specific research project you intend to pursue. Even though it is called a "personal" statement, you should nevertheless highlight your research program, though a personal statement does give you an opportunity to discuss non-research aspects of your portfolio as well.

These days, most faculty jobs ask for teaching, mentoring, and/or diversity statements, separate from the research statement. Here is advice specifically for those documents:

- **There is a wide range of opinions about how to evaluate these statements.** Most academics understand the importance of research statements and how to evaluate them. For other types of statements, though, the importance and evaluation criteria are less clear. I have served on faculty hiring committees where some of my colleagues

will merely skim teaching/mentoring/diversity statements, with the idea that the candidate can gain skills in these areas over time. But I've also talked to people who read the teaching/mentoring/diversity statement first, and only read the rest of the file if they think those statements are sufficiently mature.

- **More than any other part of your application, this is a place to "do no harm".** Given the wide range of views about these statements, it is hard to predict how this part of your application might be received. You might have a really cool way of rethinking teaching/mentoring, but if it is too far away from the mainstream, it might not resonate with your reader. Similarly, "diversity" is a hot-button topic that can be challenging to navigate even if you are well informed. You definitely want to present your viewpoints on these topics, but try to do so in a way that won't alienate a reader who might have different perspectives.
- **When in doubt, highlight your understanding of the issues.** If you've never taught in the classroom or directly supervised a student, you might not have a fully formed teaching/mentoring philosophy, but you can nevertheless emphasize some of the challenges surrounding these activities. Similarly, you might not know how to advance diversity, equity, and inclusion in academia, but you are probably aware of the problems of underrepresentation to some extent. In the spirit of "do no harm", make sure that you at least articulate your awareness of the issues surrounding teaching/mentoring/diversity and express enthusiasm about working towards positive change.
- **Specific approaches are better than generic platitudes.** In a research statement, you want to highlight your specific accomplishments and goals, such that your research statement is unique and compelling. The same goes for a teaching/mentoring/diversity statement, where you want to highlight specific things you have already accomplished and want to accomplish, rather than just generic things that anyone could say. In the spirit of "do no harm", this is not the place to present your manifesto on the ills of academia, but it is the place to articulate concrete paths for improvement.
- **Make it actionable.** Think about what "success" might look like for your teaching/mentoring/diversity activities? Are there particularly milestones or targets you want to reach? Just like your research program will have grand ambitions, your non-research activities can have long-term goals as well. The more concrete and actionable your proposal is, the more you can convince your reader that you can plausibly realize your vision.
- **Tie it back to research in some way.** One of the wonderful aspects of academia is the way that different activities can intertwine. Your teaching/mentoring/diversity activities can advance research, and vice versa. Especially when applying for positions that are research focused, building connections between your research and non-research activities can be more compelling than treating them separately. This is obvious in the case of mentoring of students/postdocs in your research group, since advancing the careers of junior people is one of the best ways to advance science. But it also worth thinking about possible ways your research could be synergistic with your teaching and diversity activities.

C.2 So You Want To... Get Strong Supporting Letters?

C.2.1 Ask Jesse for a Letter of Recommendation?

If you are reading this document, then there is a 99.9% chance I am willing to write you a letter of recommendation. The first letter of a given application season is the hardest one for me to write; each subsequent letter requires very little additional effort since the content will be nearly identical. Thus, if I agree to write you one letter, you can assume that I'm willing to write a letter on your behalf for any position.

With only rare exceptions, academic letters of recommendation are always positive (see Sec. F.1.1 where I emphasize this point to letter writers). I have never written a solicited letter where I didn't recommendation that person for the position. (External letters are a different story, but that is typically only relevant for tenure-track faculty positions.) If I can't speak to your qualifications for a position, I will simply suggest you choose a different letter writer.

But while my letters are always positive, the question is how much "ammunition" I have to advocate for you. If I say "X is a spectacular physicist", then I'd better provide ample evidence of that in my letter or else my words will read hollow. Your interactions with me directly inform my letters, which is yet another reason why you should communicate with me well and often. This is especially true during the year leading up to your application, since every interaction with me is "data" I can use to update and improve my letter for you.

To make sure I can write the most supportive letter possible, here is the information I'll need from you before I start writing. Please use my letter of recommendation request form:

- ☐ **Your résumé/CV**. Having this document in hand is important in order to make sure I don't forget to mention key facts about your experience and qualifications.
- ☐ **Your research statement (or equivalent)**. It is very bad to contradict a candidate in a letter. If you say you want to do A, but I claim you want to do B, that reflects poorly on both of us. I will align my letter to your research statement to the best of my ability.
- ☐ **A list of 3-5 key points you want me to mention in your letter**. While I likely have a good idea of your achievements, it is helpful to be reminded. And even if I have memorized your CV, these 3-5 key points can help me prioritize where I invest "ink", since I need to keep my letter to a reasonable length. Plus, you may have important achievements that I don't have first-hand knowledge of but can still discuss with authority.
- ☐ **The names of your other letter writers**. This can help ensure that the letter I write will be complementary to those of your other letter writers. Especially for letters with a word limit, it is helpful to be able to say "As you've undoubtedly heard from Prof. Y, ...".
- ☐ **Your first deadline**. As I said, the first letter is the hardest for me to write, and I want to make sure I have plenty of time (ideally a month, but at minimum two weeks) to find time to prepare it.

Of course, the earlier you give me the information above, the more likely I can give you constructive feedback on your application materials. I typically need **at least 2 weeks but ideally 4 weeks** to prepare a new letter.

It is difficult to overstate the importance of strong letters of recommendation. Letters are what get you "through the door", such that you can then be evaluated on the rest of your file. Glowing letters can't guarantee you a position, but flat letters will certainly sink your chances. In some ways, letters are the aspect of your file that you have the least control

over. But in other ways, they can be the most accurate reflection of your achievements, as long as they are written by people who know your work and are committed to your future.

- To put a fine point on it, **if I agree to write you a letter, then I am committed to your future**.

Finally, there are special cases where I might not be the best letter writer for you, even if I would be your strongest advocate. Everyone knows that an advisor will write a strong letter, so there may be rare situations where it is best for other people to trumpet your work and for me to just cheer from the sidelines.

For some positions (especially graduate school), you will be asked if you want to **waive your rights to see your letters of recommendation.** Yes, you should always waive your rights. You will get stronger letters from me (and from your other letter writers) if the letters are treated confidentially. This might seem counterintuitive, since by maintaining your right to see your letters, you can check if your letter writer wrote a positive or negative letter for you. But the reality is that I am limited in what I can say if I know that a student could eventually see what I wrote. For example, I can't include statements like "*X* is the strongest student I've worked with during my time on the MIT faculty", since that would be unfair to other students if you knew that. Letters are uniformly positive, so there is not really such a thing as a negative letter. The only question is how positive I can be, and without confidentiality, I am limited in what I can say (cf. "*X* is one of twelve students I've worked with during my time on the MIT faculty"), so the letter will end up being weaker.

C.2.2 Ask Other People for Letters of Recommendation?

Most academic positions require at least 3 letters of recommendation, with some places requiring 4. No matter how strong a letter I write for you, you won't get the job if you don't have at least 2 (or 3) other people who can advocate strongly for you. Every year, I encounter a file for a candidate with one glowing letter from their research advisor and 2 (or 3) tepid letters. Such files don't make it very far in the selection process.

Ideally, you would have more than 3 letter writers to choose from. Since some jobs limit you to exactly 3 letters, you will have to choose who your "core" letter writers are. This can be a delicate balancing act, between (at least) 3 considerations:

- **Familiarity.** The better your letter writer knows you and your work, the stronger their letter can be.
- **Seniority/Reputation.** The more experience/influence your letter writer has, the more likely they can write a persuasive letter.
- **Complementary.** The broader the range of fields and institutions represented by your letter writers, the more compelling your file will be.

In some cases, these considerations imply that I might not be the best choice for one of your core letter writers, but I might still be a good choice as a 4th (or 5th) letter to round out your application.

To get strong letters of recommendation, you need to cultivate relationships with your letter writers. There are various ways to do this:

- **Collaborate with them on a project.** The best way for someone to sing your praises is if they have first-hand knowledge of your skills.
- **Mentor their students.** Even if you don't work directly with someone, you can impress them by working closely with their mentees.

- **Present your work to them.** This can be in the context of a formal talk, journal club, or thesis committee meeting. The more someone knows what you are doing, the more they can advocate for you.
- **Help them out.** If a potential letter writer asks you to give a substitute lecture, serve on a committee, or review a grant proposal, say yes (within reason). In addition to cultivating good will, you will develop essential non-research skills for academia.
- **Make yourself known.** Sometimes all you need to do is be a visible member of the community. This is especially true if you ask for a letter from someone in a leadership role.

Of course, regardless of how you build a relationship with your letter writers, make sure to give them ample time to write your letter. No matter how much someone wants to advocate for you, they are unlikely to write a strong letter if they face an imminent deadline.

Eventually, you will get to a career stage where some of your letters will need to come from people you are at "arms length" from you. Some of the advice above still holds about making connections, but even more important is to ask your trusted mentors for advice about who to select.

C.3 So You Want To... Prepare for Job Interviews?

C.3.1 Prepare for a Virtual Interview?

Increasingly, search committees are conducting virtual interviews with candidates at multiple career stages. For graduate admissions, virtual interviews can help assess preparedness for a PhD program. For faculty searches, virtual interviews allow the committee to engage with more candidates than possible with traditional in-person visits. While the increasing prevalence of virtual interviews can be partially ascribed to positive pandemic experiences with Zoom, it also reflects a dramatic increase in the number of qualified applicants for a fixed number of positions, with virtual interviews as an additional filtering step to help narrow down a pool of top talent.

Regardless of your career stage, you will want to do ample preparation for virtual interviews. Here are some things to keep in mind:

- **You have already made the paper case.** The "long list" for virtual interviews is around twice the length of a typical "short list" for in-person evaluation. Nevertheless, the competition to secure a slot on the long list is fierce, so if you have gotten to the virtual interview stage, your CV/research statement/letters have already impressed at least one person on the search committee. While you should certainly amplify the messages you already have in your paper file, the questions you get will most likely go beyond the content of your research statement.
- **Your target audience is the most skeptical member of the search committee.** Almost certainly, the person on the search committee who is closest to your area of expertise is already convinced you should make it to the next stage, otherwise, you probably would not have secured a virtual interview slot. So your goal is to convince people outside of your core research area about your suitability for the position. Even if the questions come from committee members in your orbit, you should aim your answers toward the whole committee. Often times, the most silent member of the committee is the one whose opinion matters most.
- **Make a list of every question you can think of, and have an answer at the ready.** Imagine that you were on the search committee, and think about what questions you would ask. Typically, committees will prepare some questions that are directed to all candidates, and some questions that are candidate specific. Think about possible "softball" questions, "obvious" questions, and "unfair" questions. Search up "standard job interview questions", and think about how those might be adapted to this search. If you are worried about a specific question, then that is precisely the one that they are likely going to ask, so have an answer prepared.
- **Make a list of every key point you want to get across.** Regardless of what questions you are asked, you should try to weave in any key points you want to make. Obviously, you won't have time to say everything, but if there is something that you think would elevate your candidacy, find a way to incorporate it into the discussion. This can be delicate, since you do want to give direct answers to the questions you've been given, but there is usually a way to provide indirect answers as well and relate what you've been asked to what you really want to say.
- **Be mindful of time.** While this is hard to calibrate, try not to give overly short or overly long answers. Unless you are specifically told to say more, one to two minutes (i.e. the rough amount of time to narrate a slide in a talk) is typically the right target length for a response. This is where having a list of key points is helpful, since you can sometime add extra relevant information if you think your answer is too short. In practice, though, candidates tend to ramble on, taking valuable time away from the next question.

- **Stay positive, maintain your confidence, and don't get flustered.** A virtual interview is a test of your communication skills and adaptability under pressure. Committees often like to ask one really hard question to see if they can throw off the candidates. No matter how challenging the question might seem, don't lose your cool. This is a chance to show that you can think on your feet, and even if your answer itself isn't the best, giving an answer of some form is better than saying nothing of substance. Keep a catalog of these hard questions, both to prepare for them in your next interview, and to ask them when you are on the other side of the decision process.

C.3.2 Give a Great Job Talk?

At some level, every talk is a job talk, since you never know whether your future employer might be in the audience. But a "real" job talk requires a lot of preparation, so start early and make sure to give a practice talk in front of a live audience to get feedback. When I applied to MIT, my job talk was scheduled right after a winter conference at the Aspen Center for Physics, so I thought I was going to enjoy some skiing and relaxation in the mountains before my MIT interview. Instead, after a decidedly mediocre practice talk, I realized that I had to rework my entire talk from scratch! So no skiing for me, but the extra effort was really worth it in the long run.

In terms of specific advice, here are three typical suggestions I make at practice job talks:

- **Make sure your job talk is about you.** This might seem obvious, but sometimes applicants focus on the field as a whole and minimize their own contributions. If ever there was a time to crow about your achievements, it is your job talk! So while it is important to provide the overall context for your work, focus on what you've done and provide some indication about what you plan to do in the future. And make sure your name is prominent in the citations throughout your talk.
- **Imagine that someone only watches the first five minutes of your talk.** These days, most job talks are recorded, so it may be the case that some dean will literally only watch the first five minutes to get a sense of who you are. In those first five minutes, is it clear what you work on and broadly what you've achieved? Even if someone watches your full hour talk, the first five minutes are important to set the stage and introduce the key themes, so give a clear picture of who you are and a clear outline (and even the punchline) right at the beginning.
- **Don't bury the lede.** Academic talks shouldn't be whodunnit mysteries. If you've accomplished something, say so, say it loudly, and say it often. You might even show your most important plot right at the beginning of the talk as a preview. But, you say, isn't it more exciting to have a "reveal" at the end? It might be, if your audience is still engaged, but based on the five-minute advice above, they might not stick around for the punchline. Plus, there is still a way to pull off magic even if your audience knows exactly what is going on. If you really want a surprise ending, you can slip in a Jobsian "one more thing", but make sure your main result is front and center.

C.3.3 Interview for a Position? (♯)

C.4 So You Want To... Get Lists of Potential Positions? (♯)

C.4.1 Get a List of Potential Physics PhD Advisors?

In terms of specific places to apply, you can help yourself (and future readers of this document) by identifying potential institutions and PhD advisors here, ideally with links to the relevant webpages. The criteria for inclusion is Netflix style: "If you liked *Jesse: The Movie*, then you might also like...". You can go as broad or as narrow as you want in terms of genre. (No matter what I watch these days, Netflix is insistent that I'll enjoy *My Little Pony: A New Generation*.)

Note that appearing on this list below should not necessarily be viewed as an endorsement. You should always do your homework (including talking to previous and current students) before deciding whom to work with. A dagger (†) indicates that one of the contributors to this document has direct personal experience working with the person in question and can provide more information.

- Boston Area: *Less than 2.5-hour drive from MIT*
  - **MIT**: Tracy Slatyer†, Iain Stewart†, Jesse Thaler†, , ...
  - **Harvard**: Matthew D. Schwartz†, Matthew Reece†, Daniel Jafferis, Xi Yin, ...
  - **BU**: ...
  - **Northeastern**: ...
  - **Tufts**: ...
  - **Brandeis**: ...
  - **Brown**: Chung-I Tan, Marcus Spradlin, Jiji Fan,...
  - **UMass Amherst**: ...
- New York City Area: *Less than 2.5-hour drive from Columbia*
  - **Columbia**: Brian Greene, Norman Christ...
  - **Stony Brook / Brookhaven**: Sally Dawson,
  - **Princeton / IAS**: Igor Klebanov, Herman Verlinde...
  - **NYU**: ...
  - **Yale**:
  - **Rutgers**: ...
  - **UPenn**: Jonathan Heckman...
- Mid-Atlantic: *Maryland, Delaware, New Jersey, Pennsylvania, New York*
  - **U. Maryland**: ...
  - **Johns Hopkins**: ...

  - **U. Pittsburgh**: Brian Batell, Tao Han,...
  - **Carnegie Mellon**: ...
  - **Cornell**: ...
  - **Syracuse**: ...
  - **SUNY Buffalo**: ...
  - **U. Rochester**: ...
- Southeast: *Virginia, Florida, North Carolina, Alabama*
  - **William & Mary**: ...
  - **U. Virginia**: ...
  - **Virginia Tech**: ...
  - **U. Florida**: Wei Xue[†], Konstantin Matchev,...
  - **Florida State**: ...
  - **Duke**: ...
  - **UNC**: ...
- Chicago Area: *Less than 2.5-hour drive from Fermilab*
  - **U. Chicago / Fermilab**: David Kutasov, Savdeep Sethi, Emil Martinec
  - **Northwestern / Argonne**: ...
  - **U. Wisconsin, Madison**: ...
  - **UIUC**: Aida El-Khadra, Thomas Faulkner,
- Midwest: *Indiana, Iowa, Michigan, Minnesota, Missouri, Ohio*
  - **Ohio State**: Gregory Kilcup, Stuart Raby, ...
  - **Case Western**: ...
  - **U. Michigan**: Aaron Pierce[†], Ratindranath Akhoury, Henriett Elvang, James Liu, Leopold Zayas,...
  - **Michigan State**: ...
  - **U. Cincinnati**: ...
  - **U. Minnesota**: ...
- Southwest: *Arizona, New Mexico, Colorado, Oklahoma, Texas*
  - **U. Texas, Austin**: ...
  - **Texas A&M**: ...
  - **U. Colorado, Boulder**: Ana Hasenfratz, Oliver Dewolfe, Robert Mawhinney, ...

  - **U. Utah**: ...
  - **U. Arizona**: ...
  - **Arizona State**: ...
- San Francisco Area: *Less than 2.5-hour drive from Berkeley*
  - **U.C. Berkeley / LBNL**: Ori Ganor, Petr Horava, ...
  - **Stanford / SLAC**: Xiaoliang Qi, Stephen Shenker, ...
  - **U.C. Santa Cruz**: ...
  - **U.C. Davis**: ...
- Los Angeles Area: *Less than 2.5-hour drive from UCLA*
  - **Caltech**: ...
  - **U.C. Irvine**: ...
  - **U.C. Riverside**: ...
  - **U.C. San Diego**: ...
  - **UCLA**
  - **U.C. Santa Barbara**: Nathaniel Craig[†], ...
- Pacific: *Washington, Oregon, Hawaii*
  - **U. Washington, Seattle**: ...
  - **U. Oregon**: ...
  - **U. Hawaii**: ...
- Canada: *the land of David's*
  - **Perimeter**: (apply to McMaster/Waterloo/... and choose your supervisor here)...
  - **U. British Columbia**: Kris Sigurdson[†]
  - **TRIUMF**: (apply to UBC, SFU or UVic and choose your supervisor here): David Morrissey[†] and David McKeen[†]
  - **Simon Fraser**:
  - **UVic**: Adam Ritz, Pavel Kovtun, Kristan Jensen
  - **U. Toronto**: David Curtin, Yoni Kahn[†]
  - **McGill**: Jim Cline
  - **Queens**: Aaron Vincent and Joe Bramante
  - **Carleton**: Daniel Stolarski[†], Yue Zhang

  - **York**: Nassim Bozorgnia
  - **U Montreal**: David London
- United Kingdom:
  - **Durham / IPPP**: Djuna Croon
  - **Oxford**: ...
  - **Cambridge**: ...
- Italy:
  - **U. Padova**: ...
  - **U. Genova**: ...
- Germany:
  - **Heidelberg**
  - **Munich**
  - **Mainz**
- Japan:
  - **U. Tokyo / KIPMU**
- Israel:
  - **Technion**: ...
  - **Weizmann**: ...
  - **Tel Aviv**: ...

C.4.2 Get a List of Postdoctoral Fellowships?

Here is a list of postdoctoral fellowships relevant for theoretical particle physicists:

- **Harvard Society of Fellows**: Across all areas of the academia. (Early August deadline)
- **U.C. Berkeley Miller Fellowship**: Across all areas of the basic sciences; where I did my postdoc.
- **MIT Pappalardo Fellowship**: Across all areas of physics.
- **IAIFI Fellowship**: At the intersection of Physics and AI.
- **Simons Society of Fellows Junior Fellowship**: To work at an institution in the New York City area.
- **Princeton Center for Theoretical Science Fellowship**
- **Stanford Science Fellows**
- **Caltech Burke Fellowship**

- **UChicago Fermi Fellowship**
- **UChicago Grainger Fellowship**
- **LBNL Chamberlain Fellowship**
- **LANL Oppenheimer and Feynman Distinguished Postdoctoral Fellowships**
- **UCSB Physics Fellowship**
- **CERN Theory Fellowship**
- **Cornell Klarman Fellowship**
- **Yale Mossman Postdoctoral Fellowship**
- **Perimeter Institute Fellowships**
- **NASA Hubble and Einstein Fellowship**
- **Schmidt Science Fellows**: For switching scientific fields. (Mid-July deadline)

If you already have a foreign university in mind, you may be eligible for these fellowships:

- **Marie Skłodowska-Curie Actions Postdoctoral Fellowship**
- **Fulbright Postdoctoral Scholarship**

If you are a member of an underrepresented group or if you have significant DEI-related work on your resume, you may be eligible for these fellowships:

- **NSF MPS Ascend Fellowship**
- **U.C. Presidential Postdoctoral Fellowship Program**
- See minoritypostdoc.org for additional opportunities

Please let me know about other fellowships you are aware of so I can keep this list up to date.

Appendix D

Advice about Research Skills (♯)

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D.2 So You Want To... Present Your Work? (♯)

D.2.1 Attend Other People's Talks?

The best way to prepare for eventually giving your own talks is to attend other people's talks. This is true even if (and in some ways, especially if) you get lost within the first minute of the talk. No matter how boring or seemingly irrelevant a talk is, there is always something you can learn. Of course, you shouldn't spend all of your time going to talks, but you should get in the habit of going to talks regularly, aiming for between 2–5 per week when you are a grad student.

Here are some strategies for making the most out of a research seminar:

- **Try to identify 2–4 takeaways from the talk.** Taking extensive notes works for some people, but not for me. But I do find it helpful if I summarize a few key messages from the talk, which I am more likely to remember than specific details.
- **Practice coming up with questions.** Even if you don't ask your questions, coming up with them is a way to test your understanding and identify things you are genuinely interested to find out more. I try to come up with ~ 3 questions per talk, so that I have something to ask if the room suddenly goes silent during Q&A.
- **Identify your ignorance.** Even today, I find myself attending talks where there is some key concept that I'm completely unfamiliar with but central to the story. Of course, in the ideal case, the speaker would explain that concept clearly, but if not, then I have some homework to do to learn at least a little about it.
- **Test your knowledge.** As you experience more talks, you'll start hearing the "standard" way topics are discussed. See if you can guess the content of the next slide, especially if it is about background material you've heard before.
- **Highlight presentation tricks and pitfalls.** Almost every talk will have an "interesting" way of presenting information. I put interesting in quotes, since sometimes it works really well and sometimes not. Identify those interesting moments and form opinions about them. Borrow/steal the best strategies.
- **Suggest improvements (to yourself).** Unless it is a practice talk, most speakers won't appreciate you giving them negative feedback in the moment. But you can write down that feedback for yourself, and keep it in mind when you prepare your own talk in the future.

D.2.2 Give a Talk? (♯)

D.3 So You Want To... Promote Your Research Program? (∄)

D.3.1 Win a Prize or Award? (∄)

D.3.2 Create a Website? (∄)

D.3.3 Create a LinkedIn Profile? (∄)

D.4 So You Want To... Become an Independent Researcher? (∄)

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F.1 So You Want To... Help Your Mentees Get Jobs? (♯)

F.1.1 Write a Letter of Recommendation?

At some point in your professional career, you will be asked to write a letter of recommendation (see Sec. C.2.1 for advice from the mentee perspective). If you've never written one of these, it can be a daunting prospect, since without examples, it is hard to figure out how to calibrate your letters to the community standard. Furthermore, it is difficult to find examples to read, since these documents are typically confidential. Eventually, if you are involved in the hiring process (in academia or in industry), you will have a chance to read reference letters, but until then, you'll have to muddle your way through as best as you can. Here are some generic pieces of advice for writing letters:

- **Know if you have a page limit.** Most letters of recommendation can be any length you want, though some places impose a maximum. I tend to write 1 page for students I only know from the classroom, 2 pages for students/postdocs I have some research experience with, 3 pages for students/postdocs/faculty I have ample research experience with, and 4 pages for postdocs/faculty I am pulling out all of the stops for. If I'm on the fifth page, then I am probably over doing it so I try to cut back.
- **Err on the side of being too positive.** At least in academia, letters from North American professors tend to be quite positive in tone. (European professors are typically more muted, and it is hard for me to make a blanket statement about letters from other parts of the world.) Even negative sentiments tend to be framed in a positive light, and you rarely read outright negative letters. Of course, you should never outright lie in your letter, but know that even slightly negative statements are often interpreted as big red flags.
- **Be clear about your level of enthusiasm.** If you "recommend" someone for a position, say so. If you "strongly recommend" someone, say so. If you "enthusiastically recommend" them in the "strongest possible terms", say so, though be careful about being overly enthusiastic if you aren't actually that enthusiastic.
- **Keep in mind that your reader will try to calibrate your letter.** Most institutions keep track of all letters that they have ever received, so your letter might be compared to previous ones you've written. It is quite hard to calibrate letters across years, especially if you are writing your very first letter, but as you get more experience, you'll want to think about how your letter might be interpreted in the future. Even if the reader doesn't do a direct comparison between letters you've written, they will start to get a sense of your general letter style and calibrate accordingly.
- **Imagine that your reader only reads the first and last paragraphs.** Letters take many different forms, but the first and last paragraphs often give a summary of a candidate's qualifications and your level of enthusiasm. I try to write these two paragraphs in a self-contained way, to make sure if that if the reader only skims the rest of my letter, they will have a reasonably complete picture.
- **Back up everything you say with examples.** If you say that someone has skill X, give an example that shows that they have skill X. Examples don't have to be long (and are often better if they are short), but superlatives without substance are counterproductive. This is one of the reasons why I ask candidates to provide a list of key points they want me to highlight; see Sec. C.2.1.
- **Be clear but concise about background information.** While it is often useful to explain the relevant physics background to put your candidate's achievements in context, make sure your letter focuses on your their skills and

not on the generic state of the field. If I write more than two sentences in a row that don't mention the candidate by name, that is usually a good hint that I am spending too much time on background material.

- **Address as many qualifications as you can.** If a job requires skills X, Y, and Z, try to identify concrete examples that demonstrate that your candidate has skills X, Y, and Z (or at least is developing those skills). If you don't mention something, then your reader will assume that they don't have the corresponding skill. (As a corollary, omission is a good way to imply negative feedback without saying so outright.)
- **Use comparisons with care.** Sometimes you will be asked to rank candidates relative to their peers or give an approximate rank in their class. These comparisons are delicate and potentially biased. It is often helpful to say how many people you have worked with to roughly normalize the comparison pool, but before you make explicit comparisons, you should understand the community standards of your field.
- **Imagine your words being taken out of context.** Quoting letters is one way that letters are summarized. If you are enthusiastic about a candidate, provide a nice quotable phrase that can be pasted into a summary document. And think about whether any statement you make could be misconstrued if read in isolation.
- **Reuse your letter content.** To the extent possible, you should write generic letters of recommendation, since hyper-customization for a position can sometimes backfire. This is especially true if there are two positions at the same institution, where the same reader might read both of your letters. Your support should typically be for your candidate as a whole, not for a specific position.

One way to practice writing letters of recommendation is to provide a short blurb to someone else writing a letter. For example, I often ask a graduate student to provide one paragraph describing the achievements of a summer undergraduate researcher. This is a gentle way to get experience promoting your junior colleagues. Also, you can always show a draft of your letter to a more senior colleague to get their advice.

For second opinions about letter writing, I recommend:

- Stanford's Grant Writing Academy: "Ten simple rules for writing compelling recommendation letters"

F.2 So You Want To... Set Expectations for Your Group? (♯)

F.2.1 See Example Mentoring Expectations?

Appendix G

Advice about Technical Skills (♯)

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G.1 So You Want To... Improve Your Coding Skills? (∄)

G.1.1 Write Good Code? (∄)

G.1.2 Use a Computer Cluster? (∄)

G.1.3 Learn Machine Learning for Research? (∄)

G.2 So You Want To... Learn High Energy Physics? (♯)

G.2.1 Pass the Oral Qualifying Exam? (♯)

G.2.2 Learn Particle Phenomenology for Research with Jesse? (♯)

G.2.3 Annoy Jesse with Oversimplified Logic?

When describing the open questions in particle physics to non-experts, it is usually necessary to paint a simplified picture using simplified language. Sometimes, though, this simplified language is used by experts when communicating to each other. Mostly this is harmless, but sometimes I find that it misleads students (and even professors!) about what the real mysteries in particle physics are.

Here are two examples where I think that the statement in bold is misleading and that the more accurate statement is more interesting than the simplified lore:

- **"The Standard Model of particle physics cannot explain neutrino masses and mixings."** If by "the Standard Model", we mean a theory without right-handed neutrinos and only dimension four and smaller interactions, then this statement is true. But sometimes this statement is erroneously interpreted to imply that neutrino masses and mixing somehow expose a deep conceptual limitation to the Standard Model. I think it is more accurate to say: "The Standard Model of particle physics has many plausible extensions to accommodate neutrino masses and mixings, but we don't yet know the precise mechanism realized in nature." Viewing the Standard Model as an Effective Field Theory, the simplest mechanism to introduce neutrino masses and mixings is through the dimension-five Weinberg operator, which has the added bonus of providing a compelling scaling argument for why neutrino masses should be small but non-zero.

- **"Quantum Field Theory is incompatible with General Relativity."** If by "incompatible", we mean it can't make predictions at arbitrarily high energies or in arbitrarily curved geometries, then this statement is true. But it is not widely appreciated that the structure of General Relativity can be derived from Quantum Field Theory. The idea is to start from the assumption that the gravitational force is mediated by a massless spin-two particle and then to work out the consistency conditions for unitary quantum behavior. Satisfyingly, this derivation explains why gravity has (and must have, by quantum consistency) a universal coupling to matter. Of course, this theory breaks down if you try to use it to describe extreme quantum gravitational phenomena, but it is a perfectly adequate theory of "quantum gravity" for phenomena experienced in our day-to-day lives.

If I am on your PhD thesis committee and you make one of the above bold statements in your thesis, then there is a high probability that I'll ask you about it during your defense.

Appendix H

Advice about Soft Skills (♯)

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H.1 So You Want To... Improve Your Time Management? (♯)

H.1.1 Make a To-Do List?

I have used many different to-do systems over the years, from informal sheets of paper, to my email inbox, to OmniFocus, to Google docs. Every system has flaws, but some systems are more effective than others. And my preferences have changed over time as my responsibilities and goals change.

One thing to think about is whether your to-do list needs to be shared with anyone else. If so, sometimes a simple Google doc where you keep track of items and cross them off as you go works well. That's also a nice way of making sure your goals align with the people you are working with.

For an individual to-do list, right now (Sept 2021), I am testing out a "big picture" to-do system based on a pad of graph paper. I still have my email inbox and OmniFocus to collect information, but the graph paper helps me focus on the things that matter most for me.

Here's what I do. I use the top two sheets of tear off graph paper. Each week (ideally Friday afternoon), I remake the following pages:

- The front of page one is divided into four quadrants with lists:
  - Top 10 items (with an estimate about when I'll be able to work on them that week)
  - 11-20 items (again, with an estimate about when they will get done)
  - Watch list (things I don't need to do anything about but want to pay attention to)
  - Quick hits (small action items that take very little time)
- The back of page one is divided into four quadrants with lists:
  - Worry list (things I'm anxious about; writing them down makes me less anxious)
  - Bonus list (tasks which I can try to do if I find time, but aren't important)
  - Punt list (things I'm not going to think about for a while)
  - Long watch list (things I need to pay attention to, but far away)
- The front of page two is divided into a top banner, four quadrants, and a bottom banner:
  - Top banner: Powerful ideas for the week (meta thoughts to inspire/guide me)
  - Four quadrants: Each of my four main goals for the semester, with a thought about what I can do that week to work towards that goal, and notes about how things are going
  - Bottom banner: Surprises/lessons I learn over the week.

The key for me is not only to look at the list regularly over the day, but to revisit this list every week to reprioritize and reflect on how things are going. But you should find the system that works best for you!

H.1.2 Save Useful Information for Later Use? (♯)

H.1.3 Get People to Respond to Your Emails?

The ease of digital communication has changed the nature of work, such that we are operating more like a "hyperactive hive mind" rather than individual researchers. Tools like Slack and Zulip are often better than email, but they still perpetuate the "always on" mentality that gets in the way of creative work.

Unfortunately, email isn't going away any time soon, so we have to learn how to deal with it productively. I try to practice "inbox zero", such that I respond to emails in a timely fashion, though I'm not perfect at it. But some people have a haphazard approach to email, and may even purposely ignore emails. Here are some pieces of advice to get through to those folks:

- **Make your emails easy to respond to**. If you are looking for a yes/no answer, set up your email such that the person can just respond with "yes" or "no". If you need multiple answers, lay them out as bullets or numbered items, with a question mark for each thing you need an answer on. The corollary of this is that if you don't need an answer, don't use a question mark; as email social niceties goes, "I hope you are doing well." is better than "How are you doing?". Basically, think of emails more like lists than narratives and people will generally respond faster.
- **Avoid unnecessary back and forth.** Want to schedule a meeting? Instead of just saying "Can we schedule a meeting?", add a suggestion for concrete meeting times (and locations!) such that the recipient can respond with a simple "Yes, that time works for me." Want someone to give you feedback on a document? Send them a link to the document and also help them out by asking specific questions where you want to get feedback. Want help resolving a challenging issue? Provide information about possible paths forward that you have already considered. Email communication is asynchronous and every message takes time, so you can save time overall if you anticipate what the possible responses might be.
- **Assume everyone is acting in good faith, but understand that they are busy.** You can't expect someone to respond to a complex email request within 24 hours. But you can expect them to respond within a week, especially if it is part of their job to do so. After one week (add it to your calendar or todo list!) send a short follow-up email to remind them of your previous message. Typically, a follow-up email will be a reply (or reply all) with a one sentence reminder of some form. No need to overthink it, though sometimes it is necessary to add people to the cc list to emphasize the urgency of your request. **[I have found that a simple "Hi, I was wondering if I could follow up on this?" always works. –Rikab]** If you haven't heard within two weeks, you'll unfortunately need to find a back channel to get a response.

H.2 So You Want To... Improve Your Efficiency?

newpage

H.3 So You Want To... Improve Work/Life Balance?

H.3.1 Balance Physics Travel Versus Family Time?

H.3.2 Get Some Travel Tips? (♯)

- **Consider taking non-stop flights, even if you have to drive a few hours to your destination.** (♯)